Firm Heterogeneity and the Impact of Immigration: Evidence from German Establishments*

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Abstract

We document substantial heterogeneity in immigrant shares across employers in Germany, with large firms spending a higher share of their wage bill on immigrants than small firms. We provide evidence that fixed costs of hiring immigrants, partly policy-induced, are a key driver of this heterogeneity. We develop a model in which firms with heterogeneous productivities choose their immigrant share subject to fixed hiring costs, and show analytically how this heterogeneity affects the welfare gains from immigration. Two mechanisms are at play when firms differ in their immigrant shares. First, immigration induces native worker reallocation across firms, which affects the competition between immigrants and natives in the labor market. Second, immigration leads to larger declines in aggregate prices because firms' marginal costs are more elastic to immigrant wages, and these cost reductions are concentrated among the most productive firms. Quantitatively, ignoring within-sector heterogeneity in immigrant shares understates native welfare gains from immigration by approximately 50%. Moreover, international trade dampens these gains, with 85% of the dampening effect explained by firm heterogeneity.

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1 Introduction

Over the past two decades, the immigrant population in advanced economies has expanded by more than 80 percent, intensifying both academic and public debate over the consequences of immigration for native-born workers. A large body of research has studied this question, often implicitly assuming the existence of a representative firm. Yet firms differ along several dimensions such as size, export orientation, and workforce composition.

Using a rich dataset from Germany, we document a new dimension of firm heterogeneity: large employers are more immigrant-intensive than small employers. To the extent that immigration inflows reallocate resources across firms, differences in firms' immigrant shares may affect equilibrium adjustments in goods and labor markets. This paper asks whether and how heterogeneity in firms' immigrant shares affects our understanding of the effects of immigration on the welfare of native-born workers.

We characterize the relationship between employer size and immigrant intensity using establishment-level data from Germany between 2003 and 2011. We find a systematic relationship between employer size and immigrant intensity: the median establishment with more than 500 employees spends 5.6 percent of its wage bill on immigrants, whereas the median establishment with fewer than 10 employees spends zero. We also find evidence that fixed costs associated with hiring immigrant workers may drive this pattern. We exploit Germany's 2011 opening of its labor market to EU New Member States (NMS) as a natural experiment that lowered fixed costs of hiring these immigrants. Using an event-study approach, we show that small establishments expanded their hiring of NMS workers more than large establishments, relative to comparable workers from non-EU countries, consistent with the presence of fixed costs in hiring migrants prior to 2011.[1]

Based on this evidence, we develop a model in which firms differ in their primitive productivity and their (endogenous) immigrant share to study the general equilibrium adjust-

[1] We also find evidence suggesting that workers from different countries may have comparative advantages in performing different tasks, providing a rationale for firms' willingness to pay these costs.

ment and welfare implications of an influx of immigrants. The model features tradable and non-tradable sectors, and consumers with CES preferences over goods within each sector. Each good is produced by a firm that can employ immigrant and native workers, who are imperfect substitutes. We model the immigrant hiring decision by closely following the intermediate input sourcing literature (Blaum et al., 2018). Firms may choose to hire immigrant labor, but doing so requires paying a fixed cost for each additional source country from which they recruit. Consequently, larger firms find it profitable to recruit immigrants from more countries and to spend a larger share of their wage bill on immigrants. To fully capture the observed relationship between firm size and immigrant share, the model allows for firm-specific fixed costs of hiring immigrants, which are drawn from a joint distribution with innate productivity.

We use a simplified model to show analytically that the welfare predictions of a model that ignores the relationship between firm size and immigrant share are biased. We do so by comparing welfare gains in a model with and without heterogeneity in immigrant intensities (but with heterogeneity in productivity). First, the competition between immigrants and natives in the labor market depends on the aggregate elasticity of substitution. With homogeneous immigrant shares across firms, the aggregate elasticity of substitution coincides with the within-firm elasticity. With heterogeneous firms, however, the aggregate elasticity also reflects substitution across firms (or goods), which governs changes in firms' scale of production. When the (within-firm) substitution effect is stronger than the scale effect, immigrants crowd out natives at immigrant-intensive firms, who are reallocated toward native-intensive firms. This reallocation allows natives and immigrants to specialize in producing different goods, making the two groups less substitutable in the labor market. This, in turn, mitigates the wage pressure from immigration relative to the homogeneous case. Second, immigration generates a larger decline in aggregate prices because firms' marginal costs are more elastic with respect to immigrant wages, and these cost reductions are disproportionately concentrated among the most productive firms. Overall, the *sign of the bias* depends on whether the elasticity of substitution between immigrants and natives is larger or smaller than the elasticity of demand, while

the *magnitude of the bias* depends on the joint distribution of firm productivity and immigrant-hiring costs.

We also show analytically that immigration gains in an open economy can be larger or lower than in a closed economy due to firm heterogeneity in immigrant share. In an open economy, the larger market size amplifies the scale effect at immigrant-intensive firms, reducing the displacement of native coworkers and weakening reallocation toward native-intensive firms. At the same time, the accompanying increase in profits induces firms to pay the fixed cost to source immigrants from more countries, increasing substitution away from natives and thus reinforcing reallocation toward native-intensive firms.

We estimate the elasticity of demand from the average firms' markups, following Oberfield and Raval (2014). The (within-firm) substitution between immigrants and natives is structurally estimated using the firm's first-order condition with respect to immigrant and native labor. We regress the firm-level relative wage bill between immigrants and natives on relative employment. We extend Ottaviano and Peri (2012)'s approach to allow for heterogeneity in workers' abilities, as suggested by our model, and using a model-based firm-level instrument. At a 1% confidence level, our estimates suggest that the elasticity of substitution between immigrants and natives within firms exceeds the elasticity of demand. Thus, the observed heterogeneity in immigrant share is expected to generate larger welfare gains from immigration. We calibrate other parameters jointly to match key targeted micro- and macro-level moments in Germany between 2003 and 2011. Importantly, the observed relationship of immigrant share and firm size, our novel fact, disciplines the parameters of the joint distribution of productivities and fixed costs.

We verify that the calibrated model properly captures the mechanisms through which firm heterogeneity in immigrant share affects welfare. Our analytical results show that the heterogeneity in immigrant shares leads to heterogeneity in the elasticity of sales and the elasticity of the immigrant-to-native wage bill ratio to immigrant labor inflows. Therefore, we compare the predicted changes in firms' sales and the immigrant-to-native wage bill ratio due to an exogenous immigrant inflow between the model and the data.

To that end, we regress firm revenues and the relative wage bill ratio on the share of immigrants in the local labor market and its interaction with firm size. We use these reduced-form estimates to compute the effect of a 1% increase in the share of immigrants in the labor market for firms in different size deciles and find that the model does a good job of matching its data counterparts.

We use the calibrated model to measure the real wage (or welfare) effects of a 20% increase in the total number of immigrants, motivated by the inflow experienced by Germany between 2011 and 2017. We find that native workers in both sectors benefit from immigration despite lower wages, as domestic prices drop significantly due to lower production costs. This suggests that focusing solely on nominal wage effects may lead to opposite conclusions about immigration's welfare impact. Revenues and profits increase for both sectors, but more so in the tradable sector, where firms are more intensive in immigrant labor. Firms in the top size decile of the tradable sector, being the most immigrant-intensive, absorb a large share of incoming immigrants, leading to the reallocation of many native workers to other firms. Given their size, large firms in the non-tradable sector absorb a significant portion of these native workers. This reallocation between firms explains most of the cross-sector reallocation of native workers. Natives also reallocate within sectors toward less immigrant-intensive firms. Overall, welfare gains of native-born workers and firm owners are 0.10% and 1.23%, respectively. In monetary terms, these gains amount to $1.5 billion for native workers and $15.3 billion for firm owners.[2]

We find that the quantitative relevance of both analytical results is large in the aggregate. First, the version of our model that removes this within-industry heterogeneity in immigrant shares, while still allowing immigrant shares to vary across industries and Hicks-neutral productivity to vary across firms, predicts welfare gains roughly 50% lower. The bias is due to both a weaker drop in wages and a stronger drop in the price index in the model with heterogeneous immigrant share, with the latter explaining 80% of the

[2]These effects are relatively large. For reference, estimates of the U.S. welfare gains from China's rise in world trade range from 0.03% to 0.2% (Hsieh and Ossa (2016), Caliendo et al. (2019)).

bias. Second, in a counterfactual trade-autarky economy, welfare gains from immigration are 27% larger. Of this trade-dampening effect, 85% is accounted for by within-sector heterogeneity in firms' migrant shares.[3]

Our paper contributes to the literature on the economic effects of immigration on the receiving economy. A challenge in this literature, which is primarily empirical, is estimating immigration's impact on the *levels* of *real* wages because reduced-form coefficients do not tend to identify general equilibrium effects, and data availability limits price index computation. Some papers address this challenge by using general equilibrium models that do not require price data to quantify immigration's effects (di Giovanni et al. (2015), Desmet et al. (2018), Burstein et al. (2020), Caliendo et al. (2021), Khanna and Morales (2025), Morales (2025)). These models, unlike ours, restrict immigrant-share heterogeneity to broader units of production (e.g., industries) and do not allow it to vary across firms. By introducing heterogeneous firms that endogenously choose their immigrant intensities due to fixed costs to hire immigrants, we make three contributions. First, our results suggest that labor reallocation across firms within sectors is an important adjustment channel for aggregate production and labor markets. Second, the price effect, which tends to be empirically challenging to quantify, is important. In our setting, focusing solely on nominal wages or earnings would reverse the conclusion on whether immigration raises or lowers native welfare. Third, the aggregate elasticity of substitution does not inform welfare effects when firm heterogeneity in immigrant share arises from fixed costs.[4]

We also contribute to the stream of the empirical literature studying the effect of immigration on firms by documenting new facts on the relationship between firm size and immigration and assessing the aggregate general equilibrium effects of immigration.[5]

Our work contributes to the international trade literature studying the role of trade in

[3]The remaining 15% (0.004 p.p.) reflects mechanisms that arise even under firm homogeneity (i.e., terms of trade effects) and is remarkably similar to previous findings (Caliendo et al., 2019).

[4]Mehra and Shen (2022) and Mandelman et al. (2024) also propose models that include hiring costs to hire immigrants in the U.S. Their models are calibrated using mostly aggregate data.

[5]e.g., Kerr et al. (2015), Dustmann and Glitz (2015), Mitaritonna et al. (2017), Arellano-Bover and San (2020), Beerli et al. (2021), Egger et al. (2022), Card et al. (2023), Mahajan (2024), Gyetvay and Keita (2024), Amior and Stuhler (2024), Orefice and Peri (2024), and Mahajan et al. (2025).

shaping the effects of immigration, which dates back to Rybczynski (1955) and Samuelson (1948). Recent work quantifies the relevance of trade using modern quantitative models that abstract from firm heterogeneity (e.g., Caliendo et al. (2021) and Brinatti and Guo (2024)). We contribute to this literature by studying whether trade amplifies or dampens immigration gains due to a new mechanism, firm heterogeneity in migrant shares.[6]

This paper also contributes to the literature on the relationship between micro and macro elasticities. Existing work shows that firm heterogeneity in factor shares leads to differences between the micro and macro elasticities of substitution between factors (e.g., Satō (1975), Oberfield and Raval (2021), Lashkari et al. (2024)). This work tends to focus on aggregate variables such as the labor share rather than the implications for welfare. We link firm heterogeneity in factor shares and micro elasticities to welfare, and explain that the macro elasticity is not sufficient for welfare analysis in our fixed-cost model.

2 Data

We use a detailed, employer-employee matched dataset from Germany provided by the Institute for Employment Research (IAB). The main data source is the Longitudinal Establishment Panel (LIAB), which includes records for a large sample of establishments over the period 2003-2011.[7] The dataset contains full employment trajectories for each employee who worked at least one day for one of the establishments in the sample during the period. It also includes employee information on citizenship, occupation, education, and daily wage. While our ideal measure for immigration status would be country of birth, the data reports citizenship, where countries are grouped into ten regions. We classify as immigrants those who report non-German citizenship in at least two periods. Appendix A.1 discusses the details and robustness of this definition.

[6] Our result that the aggregate elasticity is not a sufficient statistic for the effects of immigration on native welfare relates to the stream of the trade literature studying whether firm heterogeneity matters for the welfare effects of trade (Arkolakis et al., 2012; Melitz and Redding, 2015).

[7] The data basis of this paper is the Longitudinal Model (version 1993–2014) of the Linked Employer-Employee Data from the IAB. The data were accessed on-site at the Research Data Centre of the Federal Employment Agency at the Institute for Employment Research (FDZ) and remotely.

On the establishment side, the dataset contains information on industry, location, and financials such as revenues, material use, and export share, among others. It is important to note that the data is at the establishment level, and it is not possible to link multiple establishments to a single firm. Throughout the paper, we use establishment and firm interchangeably.[8] Also, we apply survey weights to get representative aggregates whenever necessary. More information on LIAB can be found in Heining et al. (2016).

Finally, we use the Sample of Integrated Employer–Employee Data (SIEED) in Section 3.3.2 to provide evidence on the presence of fixed costs, as it includes post-2011 information and workers' country of citizenship.[9] This administrative data is a 1.5% random sample of the universe of German establishments, where we also access the entire labor market biographies of employees who worked at least one day in these establishments.

3 Firm Heterogeneity in Immigrant Share

We begin our analysis by establishing a key fact: larger establishments are more intensive on immigrant labor than smaller establishments. To assess the implications of this heterogeneity for the welfare effects of immigration on native workers, we must understand the microfoundations that determine immigrant employment shares. To do so, we begin by discarding natural potential explanations that are inconsistent with salient features of the data. Second, we show that fixed costs associated with hiring immigrant workers likely constitute the primary explanatory factor of the observed systematic relationship between employer size and immigrant intensity.[10]

3.1 Fact: Immigrant intensity increases with employer size

To illustrate the relationship between employer size and immigrant share, we classify the establishments in our sample into ten bins based on their total reported employment. For

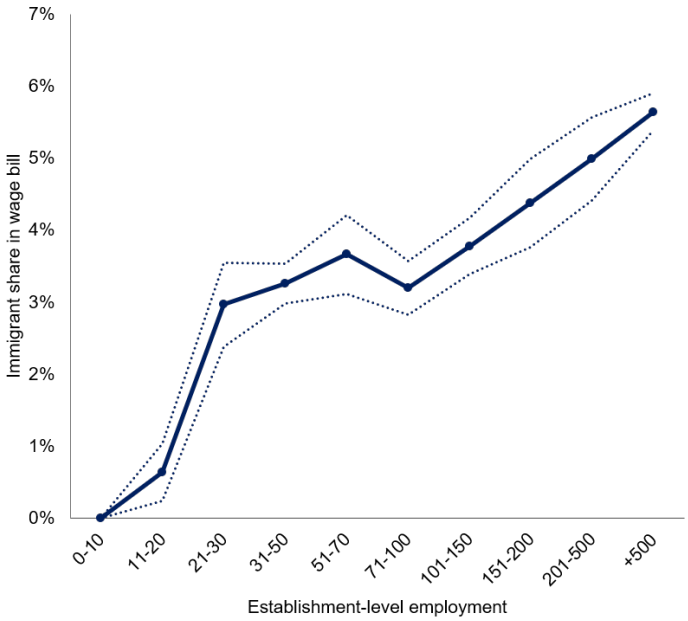
[8] Our main facts are robust to looking solely at single-establishment firms.

[9] We do not use SIEED for the rest of the analysis because it does not have firms' financial information.

[10] Appendix A.1 presents summary statistics on the sample of establishments and the distribution of immigrants across sectors and origin regions.

each bin, we plot the median share of immigrant labor in the establishment wage bill to capture the firm-level intensity of immigrants.[11] Figure 1 shows a systematic relationship between employer size and immigrant intensity. The median establishment among firms with more than 500 employees spends 5.6% of their wage bill on immigrants, while the median establishment with less than 10 employees spends zero.

Figure 1: Immigrant share across the establishment size distribution



Notes: We divide establishments into bins based on employment size and, for each bin, plot the median immigrant share of the wage bill. We compute 95% confidence intervals using 200 bootstrap repetitions. We use the years between 2003 to 2011.

3.2 Ruling out potential mechanisms

We consider potential underlying mechanisms for the documented heterogeneity. One possibility is that large employers concentrate in industries that use immigrant-supplied skills more intensively or in locations with larger immigrant networks. However, the pattern remains strong after controlling for three-digit industry fixed effects and local labor market fixed effects (Appendix Figure A1a).

Another explanation is that immigrants might reduce trade costs between origin and

[11] Our findings are robust to alternative bin definitions (e.g., deciles of revenues, or wage bill) and to using statistics other than the median, such as calculating the immigrant share for each bin, or running a regression of immigrant share on establishment size. We prefer using the median due to outliers.

destination countries (Ottaviano et al. (2018), Bonadio (2023)), which would be more relevant for exporters because they are typically larger employers. Yet, exporters and non-exporters display similar patterns (Appendix Figure A1c), suggesting that reductions in employers' trade costs are unlikely to be the main explanation.

A third explanation is due to immigrant-biased technological differences across establishments of different sizes. This explanation implies a common *elasticity* of immigrant intensity to immigration across establishments, which is counterfactual (see Section 5.5).

Appendix A.2 considers additional channels. For instance, we show that our key fact is not driven by immigrant skills and large establishments being more skill-intensive, as we find the pattern holds within skill groups. Similarly, we show our pattern is not explained by the establishment being a multinational or part of a multi-establishment firm.

3.3 Fixed costs to hire immigrants

3.3.1 Institutional aspects and descriptive evidence of fixed costs

During our period of analysis, most non-EU nationals were allowed to migrate to Germany only after securing a job offer. This arrangement, similar to the U.S. system, places the burden of navigating administrative procedures, as well as identifying and screening candidates, on the sponsoring employer (see Appendix Section A.3).

In addition to these policy-induced fixed costs, firms may incur costs required to build a recruitment channel from specific source countries. For example, firms may need to develop internal capacity to evaluate foreign credentials or learn how to access networks of suitable workers from a given origin country. This is consistent with evidence from the OECD and the German Chamber of Commerce and Industry (DIHK), who ran an employer survey in 2010 asking for the reasons why firms with unfilled vacancies did not attempt to hire workers from abroad (OECD, 2013). The top three reasons for this trend are the lack of German language skills of candidates, unclear and complex administrative procedures, and difficulties contacting candidates abroad.

The survey also shows that these barriers disproportionately impact small and medium enterprises (SMEs). One reason is that SMEs have fewer resources than large firms to overcome these barriers. For instance, larger firms are better positioned to afford specialized HR services that navigate the complexities of the immigration system and facilitate contacting candidates abroad. Moreover, the German Employment Agency verifies that the employer's petition is legitimate and that the working conditions offered to the foreign worker are not below those offered to German employees in the same occupation. According to the report, these checks tend to be more severe when the employer is not well-known, as tends to be the case for SMEs. The difficulty of SMEs to hire immigrants has even been the subject of public policy debate.

We complement this anecdotal evidence with direct evidence from the data, including the substantial share of establishments that do not hire immigrants and the lumpiness of the hiring process (see Appendix Section A.5).

3.3.2 Causal evidence of fixed costs to hire immigrants before 2011

This anecdotal and cross-sectional evidence is suggestive of fixed costs, but it does not isolate their effect from other unobserved factors that may influence immigrant hiring decisions across employers of different sizes. We make progress in this direction by using the 2011 incorporation of the New Member States (NMS) into the EU, which granted citizens from 10 countries unrestricted access to the German labor market.[12] This policy change provides a natural experiment that lowered fixed hiring costs for NMS immigrants relative to non-EU nationals, who continued to face the same restrictions. Consistent with the existence of fixed costs to hire these immigrants before 2011, we find that smaller establishments exhibited a stronger increase in NMS hiring than larger establishments.

To use time and cross-section variation introduced by this policy change, we use SIEED data because it includes several years after the policy change and citizenship country information. We estimate the effect of the policy on small and large firms' hiring of NMS

[12] Cyprus, Czech Republic, Estonia, Hungary, Latvia, Lithuania, Malta, Poland, Slovakia, and Slovenia joined the EU in 2004 but gained access to the labor market in Germany in 2011.

immigrants using the following event-study framework:[13]

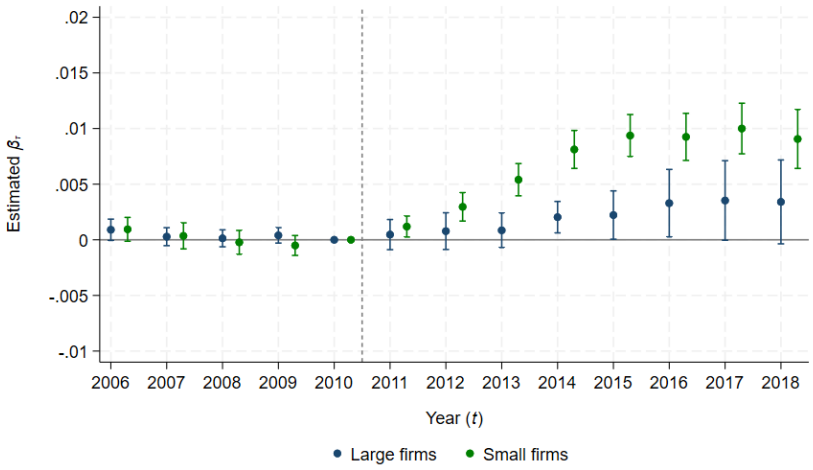
$$\frac{Emp_{o,j,t}}{Emp_{j,2010}} = \sum_{\tau \neq 2010} \left(\theta_\tau \times \mathbb{1}_j + \beta_\tau^L\right) \times \mathbb{1}_o \times \mathbb{1}_t + FE_{o,j} + FE_{j,t} + \epsilon_{o,j,t} \quad (1)$$

where $\mathbb{1}_j \equiv \mathbb{1}(j \in SME)$, $\mathbb{1}_o \equiv \mathbb{1}(o = NMS)$, and $\mathbb{1}_t \equiv \mathbb{1}(t = \tau)$. $Emp_{o,j,t}$ is the employment of nationals from group o at firm j in year t, and $Emp_{j,2010}$ is firm j's total employment in 2010, the year before the policy change. We consider two country groups: NMS members, which are the treated group, and European countries that do not yet belong to the European Union, which are the control group. SME is the group of small and medium establishments with fewer than 200 employees in 2010. $FE_{o,j}$ are fixed effects at the country group-establishment level that control for pre-existing differences between immigrant groups and establishments, such as a tendency of establishment j to hire NMS nationals. $FE_{j,t}$ are fixed effects at the establishment-year level that account in a flexible way for the effect of demand and cost shocks faced by the establishment that affect its hiring decisions. $\epsilon_{o,j,t}$ is the error term, which we cluster at the establishment level.

The coefficients β_τ^L measure the differences in the outcome variable for large firms between year t and 2010, our baseline year, for the treated group relative to the control group. Similarly, the coefficients $\beta_\tau^{SME} \equiv \theta_\tau + \beta_\tau^L$ measure the effects for small firms. For example, a positive β_τ^L for $\tau > 2010$ indicates that, on average, large establishments increased the employment of NMS nationals relative to the control group after 2010. The regression restricts the sample to firms that did not hire either treated or control group immigrants in 2005 to transparently show the effect of the policy on the extensive margin. Results are robust to not imposing this restriction, as shown in Appendix Table A4. The identifying assumption is that, conditional on the controls, the treated and control groups would have followed similar trends in the absence of the policy change. We assess the plausibility of this assumption by formally testing whether β_τ^{SME} and β_τ^L are zero before 2010. Failing to reject this hypothesis would suggest that the hiring trends for nationals

[13] Appendix Figure A8 shows the estimates of the average effect pooling together all firms, β_τ.

Figure 2: Evidence of fixed cost to hire immigrants before 2011



Notes: This figure plots the coefficients β_τ^L and β_τ^{SME} for large and small establishments from equation 1. Large establishments are firms with more than 200 employees. The sample is restricted to all establishments that did not hire NMS or other European migrants in the year 2005 (the year before our sample begins). We cluster standard errors at the establishment level and report 95% confidence intervals.

from the treated and the control group were parallel before 2010.

Figure 2 plots the estimated coefficients β_τ^L and β_τ^{SME}, suggesting that establishments began to increase their hiring of NMS nationals relative to the control group only after the restrictions on NMS nationals were lifted. More importantly, it suggests that smaller establishments increased the hiring of NMS nationals relative to large establishments.[14] We interpret the findings of this section as providing evidence of policy-induced fixed costs to hire immigrants that were in place until 2011, our period of analysis.

Robustness exercises: Given that lower-ability workers tend to work at lower productivity firms (Card et al., 2013), our estimates could be biased upward if the post-2011 inflow of immigrants had lower abilities compared to those who arrived pre-2011. However, our data suggests that, if anything, the new NMS migrants are positively selected in the post-period. For instance, the wages paid to NMS increase by 9.8% between the pre- and post-period, while they increase by 8.2% for immigrants in the control group.

[14]Moreover, column 1 of Appendix Table A3 shows that this decreasing effect on employer size persists even when the SME group is further disaggregated.

Appendix Section A.5 shows that the differential response of small and large establishments is robust to alternative specifications. For conciseness, we report the difference-in-difference estimates analogous to equation 1. Table A3 considers alternative outcome variables such as new hires, new entrants in the labor market employed by the establishment, and the immigrant share. Table A4 shows that the results are robust to using the full sample of establishments, rather than restricting to those that did not employ treated or control group immigrants in 2005, and to restricting the sample to establishments with more than 10 employees. Finally, Table A5 shows that results are robust to including an alternative fixed effects, such as origin-time fixed effects.

3.4 Additional evidence guiding modeling assumptions

For employers to optimally hire immigrants despite fixed costs, immigrants and natives must differ in the workplace. If they were perfect substitutes, firms would hire only native-born workers. Appendix A.6 shows that immigrants from different countries have different bundles of characteristics — such as age, gender, and education — and specialize in different occupations than natives. We also provide evidence consistent with this specialization being related to origin-specific comparative advantage. Based on this evidence, our model will treat immigrants and natives, as well as immigrants from different origin countries, as imperfect substitutes in production.

4 The Model

This section studies how firm heterogeneity in immigrant share shapes the economy's adjustment to immigration inflows and the effects on native workers' welfare. We develop a model informed by the evidence from previous sections. Firms with heterogeneous productivities seek to hire immigrant labor because immigrants are imperfect substitutes for native workers. However, doing so requires two types of fixed costs: an initial cost to begin hiring immigrants and an additional cost for each new country they source immigrants from. This cost structure implies that larger, more productive firms spend a

larger share of their wage bill on immigrants (Figure 1), are more likely to hire immigrants (Figure A3), and more likely to recruit immigrants from more countries (Figure A6a) compared to smaller firms. We use the model to study the role of firm heterogeneity in immigrant share in the effects of an immigrant labor supply shock. We focus on immigrant *supply* shocks to contribute to the large immigration literature studying the effects of immigrant inflows on the receiving economy.[15] We focus on the main components of the model and relegate derivations to Appendix B.

4.1 Setup

Environment: The world comprises two regions, Germany and the rest of the world (RoW), indexed by g and r respectively. The latter being composed of a continuum of countries indexed by o. In Germany, there is a tradable and a non-tradable sector indexed by $k = \{T, NT\}$, each populated by N_k firms indexed by j. Good markets are monopolistically competitive and labor markets are perfectly competitive.

Consumption: Consumers in Germany have Cobb-Douglas preferences for goods from the two sectors and CES preferences over varieties z within a k:

$$U = (Y_T)^{\alpha}(Y_{NT})^{1-\alpha} \quad , \quad Y_k = \left(\int_{J_k} (y_k(z))^{\frac{\sigma_d - 1}{\sigma_d}} dz \right)^{\frac{\sigma_d}{\sigma_d - 1}} \tag{2}$$

Y_k is the consumption of goods from k, α is the expenditure share of the tradable sector, J_k represents a fixed set of varieties available in k, and $\sigma_d > 1$ is the domestic elasticity of demand. Consumers are either firm owners, whose income is profits, or workers who earn wages. They choose $y_k(z)$ to maximize U subject to their income.

Production: Each firm produces a variety by combining native and foreign effective units of labor (d_j and x_j, respectively) in a CES manner as shown in equation 3. This modeling assumption is consistent with our evidence and follows a long tradition in the immigration literature, which understands immigrants and native-born workers as having

[15] In Section 6, however, we also study the effects of policies that reduce fixed costs to hire immigrants.

comparative advantages in different tasks (Peri and Sparber, 2009, 2011).

$$y_j = \psi_j \left(\beta_k d_j^{\frac{\epsilon-1}{\epsilon}} + (1-\beta_k) x_j^{\frac{\epsilon-1}{\epsilon}} \right)^{\frac{\epsilon}{\epsilon-1}} \tag{3}$$

where β_k is a sector-specific distributional parameter, ϵ is the elasticity of substitution between native and immigrant workers within the firm, and ψ_j is a firm-specific productivity draw. The marginal cost of production is given by equation 4:

$$u_j = \left(\beta_k^{\epsilon} w_{d,k}^{1-\epsilon} + (1-\beta_k)^{\epsilon} W_{x,j}^{1-\epsilon} \right)^{\frac{1}{1-\epsilon}} \tag{4}$$

where $w_{d,k}$ is the wage per effective unit of native labor and $W_{x,j}$ is the firm-specific wage per effective unit of immigrant labor, respectively. The share of native workers in the firms' wage bill is:

$$s_j = \frac{\beta_k^{\epsilon} w_{d,k}^{1-\epsilon}}{\beta_k^{\epsilon} w_{d,k}^{1-\epsilon} + (1-\beta_k)^{\epsilon} W_{x,j}^{1-\epsilon}} = \frac{\beta_k^{\epsilon} w_{d,k}^{1-\epsilon}}{u_j^{1-\epsilon}} \tag{5}$$

If the wage per effective unit of immigrant labor, $W_{x,j}$, were identical across firms in sector k, all firms would choose the same immigrant and native shares. However, Section 3 shows that large firms are more immigrant-intensive than small firms. To capture this fact, we need a theory based on our evidence on why firms have different s_j.

Environment to Recruit Immigrants: To theorize on the firm choice of its immigrant share that accommodates our facts and remains tractable in general equilibrium, we follow Blaum et al. (2018)'s model for intermediate input sourcing. We assume that the immigrant labor input, x_j, is a composite of labor from different origin countries:[16]

$$x_j = \left(\int_{\Sigma_j} \delta_o \, x_{j,o}^{\frac{\kappa-1}{\kappa}} \, d_o \right)^{\frac{\kappa}{\kappa-1}} \tag{6}$$

where δ_o is the productivity of workers from country o, $\kappa > 1$ is the elasticity of substitution across foreign origin countries, and Σ_j is the hiring strategy of the firm given by the countries o where the firm hires immigrants from. Given that immigrants from

[16] Appendix Section A.6 suggests that immigrants from different countries may serve as different inputs in production due to differences in comparative advantage.

different origins are imperfect substitutes, every additional origin country the firm hires immigrants from will have a positive impact on productivity and lower firm j's effective immigrant unit cost $W_{x,j}$.

Following the evidence presented in Section 3.3, we assume firms must pay a fixed cost f_k^{imm} to begin hiring immigrants from abroad and a firm-specific fixed cost f_j for each additional origin country it wants to hire from. For example, if the firm hires immigrants from two origins, it spends $w_{d,k}(f_k^{imm} + 2f_j)$ in hiring costs. One interpretation is that the fixed cost f_k^{imm} captures the costs of setting up a legal department or training HR staff in order to start hiring immigrants. The cost f_j captures the learning cost that is country-specific, such as spending resources to understand foreign education credentials and labor experience necessary to screen workers.

We make two assumptions to maintain tractability in general equilibrium. First, δ_o in equation 6 is common across firms. Firms first source from the foreign country with the largest δ_o and move down the ladder as they source from more countries. This assumption reduces the dimensionality of the sourcing problem to choosing the mass of countries $n \in [0, 1)$ to hire from. Second, we assume δ_o is a random variable distributed Pareto with shape parameter ξ and scale parameter $\bar{\delta}$. These assumptions allow us to get a closed-form expression for the wage index of immigrants as in equation 7:

$$W_{x,j} = w_{x,k}\, \bar{z}\, n_j^{-\iota} \tag{7}$$

where $w_{x,k}$ is equilibrium wage per effective unit immigrants in Germany.[17] The parameters $\bar{z} > 0$ and $\iota > 0$ are determined by $\bar{\delta}$, ξ, and κ and can be interpreted as the average productivity of immigrants and the elasticity of the immigrant unit cost to expanding the mass of sourcing countries. Intuitively, imperfect substitution of immigrants generates productivity gains from hiring immigrants from additional origins. This reduces the wage index of immigrants and the unit cost of production.

[17] In equilibrium, immigrants earn the same wage per efficient unit regardless of origin. This equilibrium result enables the closed-form solution in equation 7, which allows us to keep tractability in general equilibrium while letting efficient wages for both immigrants and natives be determined endogenously.

Distributional assumptions of ψ_j and f_j: We assume that productivity ψ_j and hiring cost f_j are jointly drawn from a multivariate sector-specific lognormal distribution with mean $[\mu_k^{\psi}, \mu_k^{f}]$, dispersion $[\sigma_k^{\psi}, \sigma_k^{f}]$, and covariance $\sigma_k^{\psi,f}$.[18]

Pricing Decision: Given CES preferences, the optimal price p_j is a constant markup over the marginal cost.

Optimal native share: An advantage of this setup is that we can write u_j, p_j, and n_j as functions of the key empirical object s_j as follows:

$$p_j = \frac{\sigma_d}{\sigma_d - 1} \frac{1}{\psi_j} \underbrace{\beta_k^{\frac{\epsilon}{1-\epsilon}} \; w_{d,k} \; s_j^{\frac{1}{\epsilon-1}}}_{u_j} \tag{8}$$

$$n(s_j) = \bar{\chi}_k \left(\frac{1}{s_j} - 1 \right)^{\frac{1}{\iota(\epsilon-1)}} \tag{9}$$

where $\bar{\chi}_k \equiv \left((\frac{1-\beta_k}{\beta_k})^{\epsilon} \; \bar{z}^{1-\epsilon} \right)^{-\frac{1}{\iota(\epsilon-1)}} (\frac{w_{x,k}}{w_{d,k}})^{\iota}$. Equation 9 follows from equations 5 and 7.

Firms maximize their profits by choosing the optimal native share s_j:

$$\max_{s_j} \; \Pi_j = \underbrace{\left(p_j(s_j) - \frac{u_j(s_j)}{\psi_j} \right) y_j}_{\text{variable profits}} - \underbrace{n_j(s_j) f_j w_{d,k} - w_{d,k} f_k^{imm} \mathbb{1}(n_j(s_j) > 0)}_{\text{Sourcing cost}} \tag{10}$$

How do firms choose their optimal native share? They face a trade-off between the drop in the marginal cost of production induced by the complementarity of hiring from an additional country and the fixed cost to source from that additional country. Given their scale of production, larger firms earn higher profits and can afford to pay f_j more times than small firms. Thus, larger firms hire immigrants from more countries than small firms, and they become more immigrant-intensive.

An important takeaway of the role of s_j in understanding the effects of immigration is the following. Firms benefit from an immigration inflow because the wage of immigrants

[18] The stricter verification procedure faced by small firms mentioned in Section 3.3 serves as a example of a negative correlation between f_j and ψ_j.

drops, and so does the unit cost of production. The size of the drop in the unit cost of production is firm-specific, and it depends on the firm's native share. In other words, the native share acts as a firm exposure to a common immigration shock and becomes the key empirical object to learn about how much each firm benefits from immigration.

Export Decision: Firms in the tradable sector can export their goods by paying a fixed cost f_x, as in Melitz (2003). The firm exports if the variable profits from export sales are larger than f_x. Under the assumption that consumers in RoW have CES preferences over tradable varieties with elasticity of demand σ_x, the optimal export price p_j^x is a constant markup over marginal cost, which now includes an iceberg cost $\tau > 1$: $p_j^x = \frac{\sigma_x}{\sigma_x - 1} \frac{u_j \tau}{\psi_j}$

Rest of the world: Given our focus on the German economy, we simplify the modeling of the RoW. We assume it has a single tradable sector, firms are equally productive, and use only native labor to produce with a constant return to scale production function $y_j^r = \bar{\psi}^r d_j^r$. Foreign firms incur trade costs τ when exporting to Germany but face no costs when exporting within the RoW.

Workers: Each country o is endowed with an exogenous mass of workers N_o born in that country. Each worker i from o draws an ability vector $\eta_{i,\ell,k}^o$ that is specific to sector k and location ℓ from a Fréchet distribution with shape parameter $\nu > 1$ and scale parameter $A_{o,k,l}$. The scale parameters are assumed to be the same across origins in RoW ($A_{o,k,\ell} = A_{x,k,\ell}$ for $o \in$ RoW), and can be interpreted as the comparative advantage of nationals from RoW and Germans in industry k in ℓ.

Labor supply to sector k in Germany: Workers from o choose the industry k and country ℓ that yield the highest real wage net of iceberg migration costs $\phi_{o,k,\ell}$ with $\phi_{\ell,k,\ell} = 1$. We assume that there is free mobility across countries within RoW and that migration costs to move to ℓ outside RoW are the same (e.g., $\phi_{o,k,\ell} = \phi_{x,k,\ell}$ for $o \in$ RoW). Due to data limitations, we also assume the cost of migration out of Germany is infinity,

such that German workers are immobile across countries.[19]

Following the properties of the Fréchet distribution, the fraction of natives and immigrants from country o who choose industry k in Germany ($\ell = g$) is:

$$\pi_{g,k,g} = \frac{A_{g,k,g}\left(w_{d,k}\right)^{\nu}}{\sum_{k} A_{g,k,g}\left(w_{d,k}\right)^{\nu}} \quad , \quad \pi_{x,k,g} = \frac{A_{x,k,g}\left(\frac{w_{x,k}}{P_g}\right)^{\nu}\phi_{x,k,g}^{-\nu}}{\sum_{k} A_{x,k,g}\left(\frac{w_{x,k}}{P_g}\right)^{\nu}\phi_{x,k,g}^{-\nu} + A_{x,r}\left(\frac{w_r}{P_r}\right)^{\nu}} \tag{11}$$

where w_r and P_r are the wage per efficient unit of labor and price index in RoW. Similar expressions apply for the share of workers from o choosing $\ell \neq g$ (see Appendix B). The share $\pi_{x,k,g}$ shows that reducing costs to migrate to Germany increases the supply of immigrants into the country.[20]

Equilibrium and Market Clearing: The equilibrium are prices, efficient wages, and labor allocations such that: workers choose the industry and destination country to work for to maximize income, consumers choose how much of each variety to purchase to maximize utility, firms choose the sourcing strategy and export status to maximize profits, all markets clear, and trade is balanced.

4.2 Firm Heterogeneity and Welfare Gains

This section shows that ignoring heterogeneity in the immigrant share across firms may lead to biased estimates of the welfare gains of immigration. To that end, we compare the analytical welfare gains of a model with heterogeneous immigrant share across firms within a sector ("heterogeneous model") and a model that ignores this heterogeneity but still allows for heterogeneity in innate productivity ("homogeneous model"). The heterogeneous model is a simplified version of our fully quantitative model, where the economy has only one sector and all firms hire some amount of immigrants (e.g., $f_k^{imm} = 0$). The homogeneous model can be a special case of this heterogeneous model with $f_j = 0$

[19] This may not be a strong assumption as the emigration rate of Germans in 2010 is approximately 3.5%, according to the IAB brain-drain dataset.

[20] The labor market in Germany is more rigid than other countries (D'Amuri et al., 2010). Such rigidity is not explicitly incorporated into our model since we allow natives to move freely across firms.

or any model following the Arkolakis et al. (2012) framework. This section assumes that the economy is in trade autarky.[21]

The native workers' welfare effects of an exogenous change in the number of immigrants, denoted by $\tilde{\mathbb{W}}^{heterog}$, is given by the following change in real wages:

$$\tilde{\mathbb{W}}^{heterog} = -\sum_{j\in J}\frac{\omega_j \tilde{s}_j}{\epsilon-1} = -\frac{\tilde{S}}{(1-\pi)\,\epsilon+\pi\,\sigma-1} = -\underbrace{\frac{\tilde{S}}{\epsilon-1}}_{\tilde{\mathbb{W}}^{homog}}\underbrace{\frac{1}{1+\frac{\sigma-\epsilon}{\epsilon-1}\,\pi}}_{Bias} \tag{12}$$

$$1-\pi \equiv \frac{\sum_{j\in J}\omega_j^S \alpha_j}{\sum_{j\in J}\omega_j \alpha_j} \in (0,1]$$

where we denote $\tilde{x} \equiv dlog(x)$, $s_j \equiv s(\psi_j, f_j)$, S is the native share in the total wage bill in the economy, $\omega_j \equiv \omega(\psi_j, s_j)$ is the share of firm j in expenditure, $\omega_j^S \equiv \omega^S(\psi_j, s_j) = \omega_j \frac{s_j}{\sum_{j\in J}\omega_j s_j}$ is the share of firm j in native employment and $\alpha_j \equiv \alpha(s_j)$ is a strictly decreasing function of s_j that determines the change of s_j to changes in the immigrant-to-native wages. The weight π depends on the heterogeneity in immigrant share and it is zero if $s_j = s$ (it occurs when $f_i \to 0$ because $\alpha_j = \alpha$).[22]

Immigration affects the welfare of native workers through two channels: a labor market competition effect and a price effect. The competition effect depends on how substitutable immigrants and natives are at the workplace, as well as how much firms expand due to the higher availability of immigrant labor. The price effect arises because immigration reduces firms' marginal costs, leading to lower prices for goods and services, which increases the purchasing power of workers' wages.

When firms employ the same immigrant share, $\pi = 0$ and the welfare gains on the right-hand side of expression 12 are given by the first component. In these models, the competition effect is characterized by an increase in immigrant shares and a drop in the marginal cost of production of the same proportion for all firms. As firms produce at

[21] Appendix B.4 discusses an alternative model where heterogeneity s_j arise from home-bias technological parameter $\beta_{k,j}$.

[22] A home-bias model provides a intuitive expression: π is proportional to the weighted variance of s_j ($\pi \propto \sum_j \omega_j (s_j - S)^2$), thus increasing with the heterogeneity in s_j (Oberfield and Raval, 2014).

lower cost and charge lower prices, the purchasing power of wages increases. The native-born real wage increase depends on the size of the inflow and the elasticity of substitution between immigrants and natives in the labor market, which is fully determined by ϵ when firms employ the same immigrant share.[23] Note that in the homogeneous model, there is no reallocation of resources across firms because they have the same cost shares.

The welfare predictions of the homogeneous model may be biased for two reasons: firms exhibit different elasticity of marginal cost and elasticity of immigrant-to-native wage bill (substitution elasticity) in response to relative wages, as shown by equations 13 and 14:

$$\tilde{u}_j = (1 - s_j)\,\tilde{W}_{x,j} = (1 - s_j)\,(1 + \zeta(s_j))\,\tilde{w}_x \tag{13}$$

$$\widetilde{(1 - s_j)} - \tilde{s}_j = -(\epsilon - 1)\,(1 + \zeta(s_j))\,\tilde{w}_x \tag{14}$$

where we normalize $w_d = 1$, $\widetilde{(1 - s_j)} - \tilde{s}_j$ is the change in relative wage bill of immigrants relative to natives at firm j, $(\tilde{W}_{x,j} + \tilde{x}_j) - (\tilde{w}_d + \tilde{d}_j)$, $1 + \zeta(s_j) \equiv \frac{\tilde{W}_{x,j}}{\tilde{w}_x} = 1 + \frac{\tilde{n}_j}{\tilde{w}_x}$ is the elasticity of firm j's immigrant bundle cost to the immigrant market wage.

To gradually build intuition on how these mechanisms affect the aggregate welfare, consider first a case where the extensive margin is not active ($\zeta(s_j) = 0$), so the heterogeneity in s_j leads to heterogeneity in the elasticity of marginal cost u_j to w_x but it does not lead to heterogeneity in the substitution elasticity.[24] In this case, the elasticity of substitution between immigrants and natives in the labor market, which we will refer to as the *aggregate* elasticity, equals $(1 - \pi)\,\epsilon + \pi\,\sigma$. The parameter ϵ measures substitution within firms, while σ captures the scale-of-production effect, namely substitution across varieties produced by firms with different immigrant intensities and resulting price declines. The difference between ϵ and σ pin down the direction of the reallocation of natives across firms with different immigrant shares.[25]

[23] The more substitutable immigrants and natives are, the lower the productivity gains for firms and the welfare gains for natives. When $\epsilon \to \infty$, the gains are zero because w_d drops in the same proportion as w_x and so does P. If ϵ is finite, $\tilde{w}_x < \tilde{P} < \tilde{w}_d$.

[24] This occurs, for example, in the home-bias model where heterogeneity in immigrant shares comes from heterogeneous $\beta_{k,j}$ as opposed to fixed costs (see Appendix B.4 for details).

[25] Mathematically, $\tilde{d}_j - \tilde{d}_{j'} = \frac{\epsilon - \sigma}{\epsilon - 1}\left(\tilde{s}_{dj} - \tilde{s}_{dj'}\right) \approx (\epsilon - \sigma)(s_{j'} - s_j)\tilde{w}_x$

In the edge case of $\epsilon = \sigma$, the substitution and scale effects cancel out, and immigrants do not crowd-in or crowd-out native workers. Given that the reallocation of natives across firms is muted, the demand response for native labor and welfare gains is the same as those predicted by the homogeneous model. When the elasticity of substitution within the firm is stronger than the elasticity of demand ($\epsilon > \sigma$), immigrants crowd out natives from immigrant-intensive firms, and natives are reallocated toward native-intensive firms. Such an increase in specialization between natives and immigrants in producing different varieties makes them less substitutable in the labor market than when natives do not reallocate across firms. Given that this reallocation adjustment is absent if firms employ the same immigrant share, the increase in the aggregate demand for natives and welfare is larger in the heterogeneous world. Similarly, when $\epsilon < \sigma$, welfare gains are lower in the heterogeneous world.

Now, consider the case where firms with different immigrant shares adjust the number of sourcing countries differently. The increase in profits following the drop in w_x induces firms to pay more times the fixed cost to source additional immigrants from more origins, which further reduces the cost of production. Therefore, p_j is more elastic to w_x and so is the price index. The resulting decline in the price index is larger when cost reductions are concentrated among large firms, which have a disproportionate weight in consumers' consumption baskets.

To sum up, equation 12 shows that the sign of the bias depends on the sign of $\epsilon - \sigma$, and its magnitude also depends on the correlation between firm size and immigrant share. Given this insight, we will formally test whether $\epsilon - \sigma > 0$, and calibrate the model to match the observed correlation between immigrant share and firm size. We will also validate the quantitative model based on the insights from equation 13 and 14, which highlights the role of the heterogeneous response of u_j (or equivalently sales) and immigrant-to-native wage bill ratio, respectively.

We conclude this section by briefly discussing the implications of our results for the relationship between micro and macro elasticities, and their consequences for welfare.

When firm heterogeneity in factor shares does not lead to firm heterogeneity in elasticity of substitution between factors of production, the weighted average in equation 12 coincides with the aggregate elasticity, making the aggregate elasticity sufficient for inferring welfare gains. However, in our fixed-cost model, where factor share heterogeneity also leads to heterogeneous substitution elasticities, the two diverge. As a result, the aggregate elasticity is not informative about welfare effects because it does not fully capture the first-order impact of price changes. Micro elasticities and firms' shares are needed for welfare analysis.[26]

4.3 Heterogeneity in immigrant share and international trade

We now ask what this firm heterogeneity in immigrant shares implies for the characteristics of countries that make immigrant flows more valuable. In particular, motivated by fundamental theorems of international trade (Samuelson (1948), Rybczynski (1955)), we study if the gains from immigration are larger or lower when a country is open to trade *due to migrant share heterogeneity.*

We compare the welfare gains from immigration in an economy open to trade and the corresponding one under trade autarky. To simplify the analytical expressions, we focus on an open economy where trade is balanced, $\sigma_d = \sigma_x = \sigma$, and firms must pay a fixed cost to export and iceberg trade costs. The welfare gains are given by equation 15:

$$\tilde{\mathbb{W}}^{heterog,open} = -\lambda \frac{\tilde{S}}{(1-\pi^{open})\epsilon + \pi^{open}\sigma - 1} - (1-\lambda)(\tilde{w}_r - \tilde{w}_d) \tag{15}$$

where λ is the aggregate domestic trade share, and π^{open} is defined as before but evaluated in the equilibrium of the open economy. The welfare effects under trade autarky in equation 12 are a special case of equation 15 where $\lambda = 1$ and $\pi(\{s_j, \omega_j\})$ is evaluated in the equilibrium under trade autarky. Throughout this section, we relabel $\tilde{\mathbb{W}}^{heterog}$, $\tilde{\mathbb{W}}^{homog}$, and π in equation 12 as $\tilde{\mathbb{W}}^{heterog,closed}$, $\tilde{\mathbb{W}}^{homog,closed}$, and π^{closed} to distinguish them from their open-economy counterparts.

[26] Mathematically, $\tilde{\mathbb{W}}^{heterog} = -\frac{\tilde{S}}{\epsilon^{agg}-1}\left(\sum_j \omega_j \frac{1-s_j}{1-S}\zeta_j\right)$ where ϵ^{agg} is the aggregate elasticity.

The analytical expression reveals two differences in the welfare effects of immigration between an open and a closed economy. First, the second term on the right-hand side vanishes in the closed economy because λ equals one. When workers reallocate from the RoW to Germany, labor costs abroad increase relative to native-born wages in Germany ($\tilde{w}_r - \tilde{w}_d > 0$). In an open economy, this reduces the purchasing power of Germans by increasing the price of imported goods, an effect that does not occur under trade autarky. This terms-of-trade-type effect operates even when firms have the same immigrant share and is present in standard quantitative models of international trade and migration.

Second, the value of π in the open economy can be larger or smaller than under trade autarky. This effect operates only if firms differ in the immigrant share; otherwise $\pi = 0$ in an open and closed economy. This is the novel mechanism affecting the role of trade in the wage effects of immigration. Intuitively, the strength of the reallocation of natives toward native intensive firms can be weaker or stronger in an open economy than in a closed economy. To see this, consider two firms: firm j, which is relatively productive and exports when the country is open to trade, and firm j', which is relatively unproductive and does not export. Because more productive firms are more immigrant-intensive, an inflow of immigrants leads firm j to expand production relative to firm j'; more so when the economy is open to trade due to the larger market size for exporters. On the one hand, this stronger increase in the scale of production at the immigrant-intensive firms in an open economy leads immigrants at that firm to displace fewer native coworkers, and reallocation toward native-intensive firms is weaker. On the other hand, the associated stronger increase in profit in the immigrant-intensive firms is larger, so firms source immigrants from more countries, replacing more natives at the firm, and strengthening the reallocation toward native-intensive firms.

According to equation 15, the difference between the welfare gains in an economy with homogeneous immigrant share ($\pi = 0$), with and without trade, is informative about the

first mechanism:

$$\tilde{\mathbb{W}}^{homog,open} - \tilde{\mathbb{W}}^{homog,closed} = (1-\lambda)\,\frac{\tilde{S}}{\epsilon-1} - (1-\lambda)\,(\tilde{w}_r - \tilde{w}_d)$$

The extra difference between the gains in an open relative to a closed economy when firms employ different immigrant shares is informative about the second mechanism:

$$\begin{aligned}\left(\tilde{\mathbb{W}}^{heterog,open} - \tilde{\mathbb{W}}^{heterog,closed}\right) - \left(\tilde{\mathbb{W}}^{homog,open} - \tilde{\mathbb{W}}^{homog,closed}\right) = \\ \frac{\tilde{S}}{\epsilon-1}\,(\sigma-\epsilon)\left(\lambda\frac{\pi^{open}}{(1-\pi^{open})\epsilon+\pi^{open}\sigma-1} - \frac{\pi^{closed}}{(1-\pi^{closed})\epsilon+\pi^{closed}\sigma-1}\right)\end{aligned} \tag{16}$$

This expression shows how firm heterogeneity in immigrant share matters for the role of international trade in the gains from immigration. In Section 6.3, we quantify its quantitative relevance by computing the left-hand side.

5 Model Estimation and Validation

We calibrate the parameters of the model in two steps. First, we estimate outside the model the elasticity of demand in Germany and the rest of the world (σ_d and σ_x), the elasticity of substitution between immigrants and natives (ϵ), the elasticity of substitution across origin countries (κ), and the elasticity of labor supply (ν). Second, we calibrate the remaining parameters to match relevant aspects of the data. Finally, Section 5.5 verifies that the calibrated model properly captures the mechanisms through which firm heterogeneity in immigrant share affects welfare by comparing predicted model responses with those observed in the data.

5.1 Elasticity of Demand

Following Oberfield and Raval (2014), we infer the demand elasticity σ_d from firms' markups, i.e., the ratio of revenue to total costs. According to the model, the following moment condition holds:

$$\mathbb{E}\left(\frac{Revenue_j}{Cost_j}\right) = \frac{\sigma_d}{\sigma_d - 1} \tag{17}$$

where j are firms in the non-tradable sector. We compute the average revenue-to-cost ratio across firms and use equation 17 to infer the value of σ_d.[27] The average markup for firms in the non-tradable sector is 1.47, which implies that $\hat{\sigma}_d = 3.1$.

Based on the analogous model's moment condition for firms in the tradable sector, we calibrate the demand elasticity from the RoW σ_x, to match the average revenue-to-cost ratio of firms in the tradable sector, given the calibrated value of σ_d and the average share of exports in their sales.[28] We find that $\hat{\sigma}_x = 4.3$, which implies an average elasticity in the tradable sector of 3.44.

The calibrated values are consistent with previous estimates in the literature, which vary by the granularity of sectoral definitions. For instance, for narrower categories than ours, Broda and Weinstein (2006) found an average elasticity of 4, and for broader categories, estimates tend to be around 0.5 (Cravino and Sotelo (2019), Comin et al. (2021)).

5.2 Elasticity of Substitution Between Natives and Immigrants

We use the first-order conditions with respect to d_j and x_j to obtain equation 18.

$$Log\left(\frac{\text{Wage Bill Immig}_{j,t}}{\text{Wage Bill Natives}_{j,t}}\right) = \frac{\epsilon - 1}{\epsilon} Log\left(\frac{\hat{x}_{j,t}}{N_{g,j,t}}\right) + FE_j + FE_{m,t} + FE_{k,t} + u_{j,t} \quad (18)$$

where $N_{g,j,t}$ is the number of natives employed at the firm at year t and $\hat{x}_{j,t}$ is the immigrant effective units CES composite. We incorporate a rich set of fixed effects FE_j, $FE_{m,t}$, and $FE_{k,t}$ which are firm, commuting zone-year and industry-year fixed effects, respectively. We denote the error term $u_{j,t}$.

In Appendix C.1, we explain in detail how we go from the first order conditions to estimating equation 18. A key part of the process is that we use the Fréchet assumptions for the

[27] Rather than using variable costs in the calibration of σ_d, which is unobservable, we use total costs (the sum of wage bill and material bill), which include fixed costs. As a result, our estimate could be upward-biased. Note that the direction of the bias means that it is even more likely that $\sigma < \epsilon$. Quantitatively, we expect its effect to be small, as the wage bill of occupations related to fixed costs accounts for approximately 1% on average of our variable costs.

[28] We use the following condition $\mathbb{E}\left(\frac{Cost_j}{Rev_j}\right) = \frac{Rev_d}{Rev}\frac{Cost_d}{Rev_d} + \frac{Rev_x}{Rev}\frac{Cost_x}{Rev_x} = \frac{Rev_d}{Rev}\frac{\sigma_d - 1}{\sigma_d} + \frac{Rev_x}{Rev}\frac{\sigma_x - 1}{\sigma_x}$

ability draws to separate ability units from labor units. We then follow the approach by Ottaviano and Peri (2012) to construct the CES composite of immigrant effective units. We use a bootstrap procedure to generate confidence intervals that properly account for the fact that $\hat{x}_{j,t}$ arose from a previous estimation procedure.

There are two concerns to identify $\frac{\epsilon-1}{\epsilon}$ by estimating equation 18 via ordinary least squares approach (OLS). First, measurement error in $x_{j,t}$ may correlate with firm, industry, or labor market factors determining hiring decisions. Second, omitted variable bias would arise if unobserved shocks to a firm's demand, technology, or shocks to a firm's labor market induce firms to expand and hire more immigrants *relative* to natives.

To address concerns not captured by the fixed effects above, we use an instrumental variables approach. We construct our instrument, denoted by $Z_{j,t}$, following the model's definition of $dLog(\frac{x_{j,t}}{N_{g,j,t}})$:[29]

$$Z_{j,t} = \sum_{o \neq g} \left(\frac{\text{Wage Bill Imm}_{o,j,2003}}{\text{Wage Bill Imm}_{j,2003}} \right) \frac{\Xi_{o,t}}{\Xi_{g,t}} \tag{19}$$

where $\Xi_{o,t}$ is the national employment growth from country group o between 2003 and t. This is a shift-share instrument where the shift is given by the growth of immigrants from nationality o relative to the growth of native workers, and the share is given by the fraction of immigrants of each firm that is accounted for by nationals from o.

Our 2SLS estimate is 0.87, and the F-stat of the first stage is 78.7. This result implies an estimate of ϵ is 7.95 (CI between 4.33 and 29.58).[30] Appendix Section C.1 includes the OLS and 2SLS results, and Appendix Section C.3 includes pre-trend tests and diagnostics on the validity of our shift-share design (Goldsmith-Pinkham et al., 2020).

[29] i.e., $dLog\left(\frac{\hat{x}_{j,t}}{N_{d,j,t}}\right) = \sum_{o \neq g} \left(\frac{\text{Wage Bill Imm}_{o,j,t}}{\text{Wage Bill Imm}_{j,t}}\right) dLog\left(\frac{N_{o,t}}{N_{g,t}}\right)$, where $N_{c,t} \equiv \sum_{j} N_{c,j,t}$.

[30] The elasticity of substitution among workers using a CES aggregator has been estimated in various studies, with notable papers reporting values ranging from 4 to 20 (Burstein et al., 2020; Cortes, 2008; Ottaviano and Peri, 2012). While our estimate is within the range of this literature, differences in the level at which production is defined, the nesting order, and the composition of the immigration shock make direct comparisons challenging. Mahajan et al. (2025) estimate the elasticity of substitution between college-educated immigrants and natives within firms in the U.S. to be 4.3.

5.3 Test for the sign of $\epsilon - \sigma$

Section 4.2 explains that the welfare gains predicted by the homogeneous model are downward biased if $\epsilon > \sigma$ and upward biased if $\epsilon < \sigma$. Therefore, we formally test the null hypothesis that $\epsilon - \sigma < 0$. We use 5,000 bootstrapped samples to obtain the empirical distribution for the statistic $\widehat{\epsilon - \sigma}$. We find that $\widehat{\epsilon - \sigma} < 0$ in only 0.007% of the 5000 cases. That is, the p-value of the null hypothesis that $\epsilon - \sigma < 0$ is 0.007, suggesting that the welfare gains of native workers predicted by the homogeneous model might be downward biased. Appendix C.2 provides details and the empirical distribution.

5.4 Additional Parameters

This section briefly discusses the calibration of the remaining structural parameters. See Appendix C.4 for more details.

The elasticity of labor supply ν: We estimate ν to be 6.14 using the residual variance of wages following the approach used by Hsieh et al. (2019) and Fan (2019).

Cobb-Douglas parameter α: We set $\alpha = 0.68$ to match the share of the tradable and non-tradable expenditure of German consumers (World Input-Output Tables).

Indirect inference approach: The remaining fourteen parameters are jointly calibrated by minimizing the distance between fourteen model-simulated moments and their empirical counterparts. Table 1 reports the calibrated parameter values, and Appendix Table C7 compares the moments in the data and the model.

Although all parameters are estimated simultaneously, each parameter is primarily informed by a subset of moments. Moments of the joint distribution of firm size and its immigrant share are informative about the parameters of the joint distribution of fixed costs f_j and productivity ψ_j. In particular, the variance of log revenues conditional on the immigrant share in sector k (and on exporter status in the tradable sector) is used to identify the dispersion of productivities $\sigma_{\psi,k}$. The standard deviation of the log immigrant-to-native wage bill ratio in sector k identifies the variability of fixed costs $\sigma_{f,k}$.

The difference in the mean of $1 - s_j$ between firms in percentile 90 relative to percentile 50 is used to identify the correlation between productivities and fixed costs $\sigma_{\psi,f,k}$.[31]

For the remaining parameters, we use the aggregate immigrant share by sector to identify β_k, the distributional share parameter in the production function. The fraction of firms that hire immigrants helps identify the base fixed hiring costs f_k^{imm}. The average immigrant share across all firms and sectors is used to identify ι, the elasticity of $W_{x,j}$ to the mass of countries the firm sources immigrants from. We use the mean ratio of export to domestic revenues for exporters and the fraction of firms that export in the tradable sector to identify the iceberg cost τ and the fixed cost of exporting f_x, respectively. Finally, we use the relative GDP per capita between Germany and the RoW to identify the mean productivity of RoW $\bar{\psi}^r$.

Table 1: Parameter Calibrated to Match Micro- and Macro- Moments

Parameter description	Parameter	Estimate	Parameter description	Parameter	Estimate
Share of natives, T	β_T	0.89	Covariance of ψ and f_j, NT	$\sigma_{\psi,f,NT}$	30.82
Share of natives, NT	β_{NT}	0.88	Fixed cost of immigrants, T	$f_{imm,T}$	2.79E-05
Dispersion in ψ_j, T	$\sigma_{\psi,T}$	1.15	Fixed cost of immigrants, NT	$f_{imm,NT}$	0.0008
Dispersion in ψ_j, NT	$\sigma_{\psi,NT}$	0.37	Productivity in RoW	ψ_x	1.53
Dispersion in f_j, T	$\sigma_{f,T}$	15207	Fixed cost of exporting	f_g	0.003
Dispersion in f_j, NT	$\sigma_{f,NT}$	16385	Iceberg trade cost	τ	1.35
Covariance of ψ and f_j, T	$\sigma_{\psi,f,T}$	-54.03	Elasticity s_j to n	ι	0.002

5.5 Model Validation: Heterogeneous Response by Firm Size

Before using the model for counterfactual analysis, we must verify that the calibrated model properly captures the main mechanisms through which immigrant-share heterogeneity affects welfare. Specifically, this heterogeneity in immigrant shares matters because it leads to heterogeneity in the elasticity of unit production costs or, equivalently,

[31] Appendix Figure A1b shows that the correlation is much weaker in the non-tradable sector than in the tradable. With $\sigma_{\psi,f,k} = 0$, the model predicts a positive size–immigrant share correlation, but one that is too weak in the tradable sector and too strong in the non-tradable sector. Thus $\sigma_{\psi,f,T}$ must be negative and $\sigma_{\psi,f,NT}$ positive to match the data.

sales (equation 13) and the elasticity of the immigrant-to-native wage bill (equation 14) to immigrant labor inflows. Therefore, we validate the model by comparing its predicted changes in firms' sales and the immigrant-to-native wage bill ratio due to an exogenous immigrant inflow with those empirically estimated from the data.

In the model, the change in the outcome variables is only due to the exogenous inflow of immigrants. On the contrary, in the data, firms' outcomes are affected by many factors. We isolate the effects of an exogenous immigration inflow from other factors that are absent in our model by estimating equation 20:

$$y_{j,t} = \theta_1 \, I_{m,t} \; + \theta_2 \, I_{m,t} \, log(emp_{j,2003}) \; + \theta_3 \, X_{j,t} \; + \delta_j \; + \delta_{k,t} \; + \delta_m t \; + u_{j,t} \qquad (20)$$

where $y_{j,t}$ is either log of sales or immigrant-to-native wage bill ratio of establishment j in year t. Sub-indices m and k denote the labor market, defined by commuting zones, and 2-digit industries, respectively. $I_{m,t}$ is the share of immigrants in the total wage bill of the labor market, and $emp_{j,2003}$ captures the initial establishment size. $X_{j,t}$ is j's initial cost-to-revenue ratio interacted with year dummies, to compare firms with similar mark-ups, and initial employment size interacted with a time trend, which increases the precision of our estimates. In terms of fixed effects, this specification allows for labor markets to be in different linear trends as captured by $\delta_m t$, for factors affecting all establishments in an industry over time captured by $\delta_{k,t}$, and for time-invariant establishment unobservable characteristics δ_j that may confound the effects of immigration. We cluster the error term ($u_{j,t}$) at the industry level and labor market level.

We estimate equation 20 via 2SLS using a shift-share instrument inspired by Card (2001) and Ottaviano et al. (2018). The instrument interacts the labor market share of the wage bill of each origin country in 2003 with a time-shifter that captures the national growth rate, from 2003 to year t, of immigrants from origin o relative to the working-age population growth in Germany. Thus, this shift-share instrument interacts country-specific flows of migration with their initial differential presence in local labor markets in Germany. The interaction term is instrumented by this labor market instrument

interacted with the initial size of the firm, $log(emp_{j,2003})$.

This section focuses on the comparison between the data and model responses, and relegates the implementation details and regression tables to Appendix C.5. For the sake of the economic interpretation, we compute the elasticity or semi-elasticity of $y_{j,t}$ to $I_{m,t}$, denoted as $\epsilon^{y}_{j,t}$, as in equation 21. When y is the log of sales, $\epsilon^{y}_{j,t}$ equals and elasticity of y, and when y is the wage bill ratio, it equals the semi-elasticity.

$$\epsilon^{y}_{j,t} \equiv \Big(\hat{\theta}_1 \ + \hat{\theta}_2 \ log(emp_{j,2003})\Big) I_{m,t} \tag{21}$$

Using the calibrated model, we generate a marginal change in the immigrant share in each sector and compute the change in the two outcome variables of interest.[32]

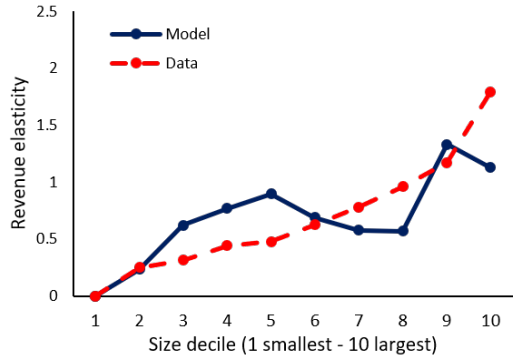
With this model-generated data and the empirical estimates from equation 21, we construct Figure 3, treating both datasets in the same way. Specifically, we plot the average effect across all firms within each decile, separately by sector. While the *levels* of the responses between the data and the model are not directly comparable, the *relative* differences across deciles are. The levels are not comparable because the estimates of the empirical regression identify relative effects, but might not identify the level effect of the shock (see Nakamura and Steinsson (2014); Wolf (2023) for a discussion on the "The Missing Intercept Problem"). To compare the relative differences, we normalize both the data and model responses to be zero in the first decile and rescale the normalized values as percentage changes with respect to their own average.[33] Figure 3 shows that, in both sectors, the model-predicted changes closely match the revenue and relative immigrant-to-native wage-bill responses observed in the data. We therefore interpret the model's performance as successful in matching these untargeted moments, which are crucial for the welfare effect of immigration. We interpret the model's performance as successful in matching these untargeted moments of the data.

[32] We lower migration costs such that the total number of immigrants increases by 1% in each sector.

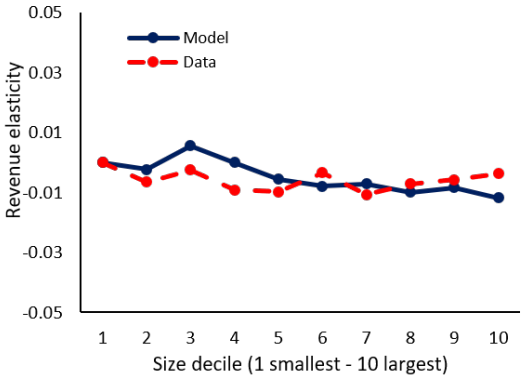
[33] We rescale the normalized values by multiplying them by the average response in the data across deciles relative to the average response in the model.

Figure 3: Immigration effects by Firm Size: Data vs. Model

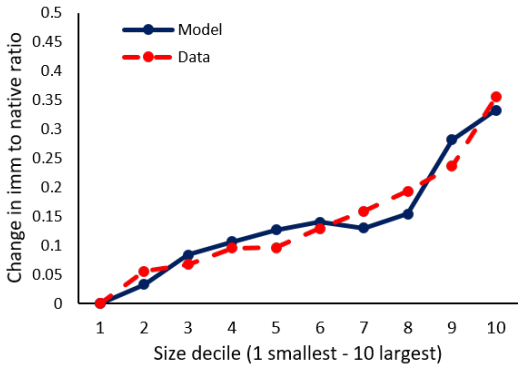
(a) Revenues - Tradable sector



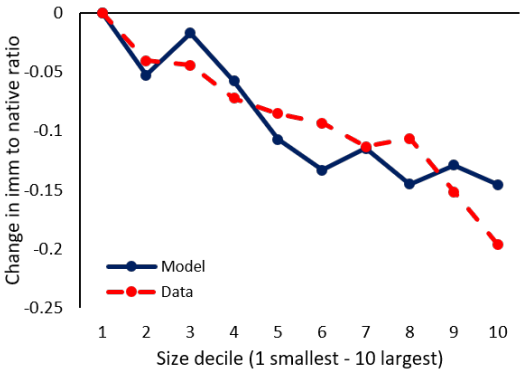
(b) Revenues - Non-Tradable sector



(c) Relative Wage Bill - Tradable sector



(d) Relative Wage Bill - Non-Tradable sector



Notes: We rank establishments into size-based deciles, with decile 1 being the smallest. We compute the elasticity of revenues (top panels) and immigrant-to-native wage bill ratio (bottom panels) to a 1% increase in the aggregate immigrant share and calculate the mean elasticity across firms in each decile in each sector. We normalize both the data and model responses to be zero in the first decile and rescale the normalized values as percentage changes with respect to their own average.

6 Aggregate implications

Having validated our calibrated model, we use it to study several counterfactuals. Section 6.1 quantifies the economic and welfare effects of an inflow of immigrants into Germany. Section 6.2 quantifies the importance of accounting for firm heterogeneity for these welfare effects. Section 6.3 studies how openness to trade affects the gains from immigration through firm heterogeneity in migrant shares. Finally, Section 6.4 evaluates the aggregate effects of alternative policies that reduce the fixed costs of hiring for specific firms.

6.1 Aggregate adjustment to an exogenous inflow of immigrants

We use the model to calculate a new equilibrium, where the total number of immigrants in Germany increases exogenously by 20%. The size of the shock mimics the magnitude of the immigration wave that occurred in Germany between 2011 and 2017. According to the OECD, the total number of immigrants in Germany went from 10.55 million in 2011 to 12.74 million in 2017, a 20.7% increase.[34]

We compute the welfare of native-born workers as their real labor income. Table 2 shows that the welfare of native workers would increase by 0.1%, representing $1.5 billion for the aggregate economy.[35] These gains are fully explained by the drop in the cost of the consumption basket: while nominal wages decrease by 0.06%, prices decrease by 0.16%. Wages decrease because the expansion in production does not fully offset the substitution effect between natives and immigrants, leading to an inward shift in aggregate native labor demand. The positive welfare effects alongside negative wage effects highlight the importance of accounting for changes in the price index when evaluating the impact of immigration, which tends to be absent in empirical work.[36] The welfare gains of firm owners (1.23%) are larger than those of native workers because they face the same price decreases but do not compete with immigrants in the labor market. Total income, which includes earnings for all workers and profits, increases by 1.21% or $16.8 billion.

Figure 4 shows the reallocation of natives across firms, which illustrates the intuition laid out in Section 4.2. Firms in the top size decile of the tradable sector, being the most immigrant-intensive, absorb a large share of incoming immigrants, leading to the reallocation of many native workers to other firms (approximately 30,000 in total). Given their size, large firms in the non-tradable sector absorb a significant portion of these native workers. This reallocation between firms explains most of the cross-sector reallocation

[34] We reduce the migration cost from the RoW to Germany, $\phi_{x,k,g}$, such that it increases the total stock of immigrants by 20% in each sector. This counterfactual does not aim to evaluate specific post-2011 migration events, such as the EU enlargement or the Syrian refugee crisis.

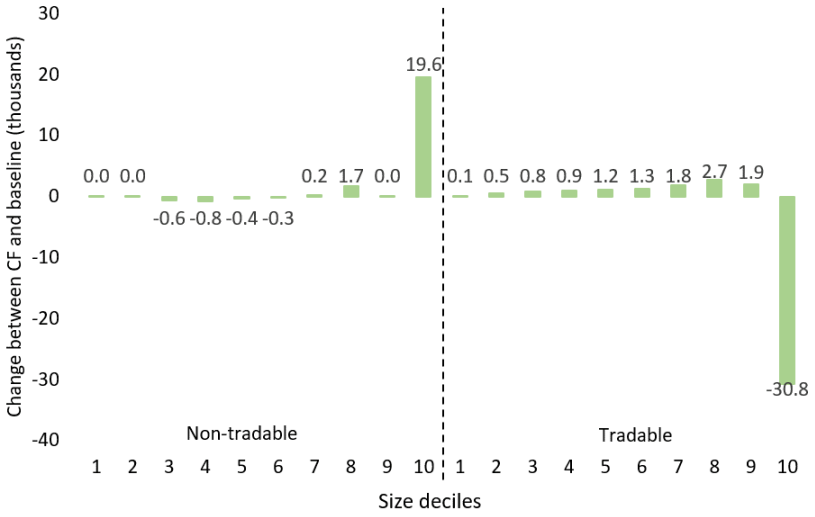
[35] These effects are relatively large. For reference, estimates of the U.S. welfare gains from China's rise in world trade are of a similar order of magnitude (Hsieh and Ossa (2016), Caliendo et al. (2019)). Monetary gains use average wages, PPP-adjusted at 2019 dollars, and total workforce from the OECD.

[36] A challenge of this literature is estimating the real wage effects of immigration because data availability limits price index computation.

of native workers. Natives also reallocate within sectors toward less immigrant-intensive firms. As previously noted, some small firms in the non-tradable sector with high immigrant shares also displace natives in response to immigration (deciles 3–6), but given their low market shares, the overall effect of this reallocation is limited.

Wages per native worker decline in both sectors, partly because wages per effective unit of native labor fall as the substitution effects of immigration outweigh the scale effect (see Appendix Table D1). In the non-tradable sector, the wage drop is also driven by a selection effect: as lower-ability natives move from the tradable to the non-tradable sector, the average ability of workers in the non-tradable sector falls, while it rises in the tradable sector.[37]

Figure 4: Reallocation of natives across sectors and firms.



Notes: The x-axis shows the distribution of firms in the non-tradable (left) and tradable (right) sectors in terms of deciles of revenues. The bars plot the absolute change in thousands of workers employed in each decile between the baseline and the counterfactual scenario of a 20% increase in the number of immigrants in each sector.

6.2 The Role of Immigrant Share Heterogeneity on Welfare

This section quantifies the importance of the documented heterogeneity in how immigration affects native workers' welfare. As suggested by the analytical results in Section 4.2,

[37]These findings contrast with Rybczynski (1955)'s theorem, whose assumptions imply that factor prices and factor shares are fixed. Under those conditions, economies absorb cross-country labor reallocation solely through output adjustments: production expands in the immigrant-intensive sector and contracts in the native-intensive one, which requires natives to move from the latter to the former.

one way to do so is by comparing the effects of an exogenous increase in labor supply $\tilde{S}$ across two models: the *heterogeneous model*, our full-fledged baseline model without the restrictions imposed on Section 4.2, and the *homogeneous model*, which does not have within-industry firm heterogeneity in immigrant shares.[38] Note that the homogeneous model allows immigrant shares to differ across sectors and size to differ across firms within industry. Specifically, in terms of primitives, the homogeneous model differs from the heterogeneous model in that fixed costs to hire immigrants are zero ($f_j = f_k^{imm} = 0$), while the innate firm productivity draws ψ_j, β_k and other parameters remain the same. Thus, the comparison across models is informative about the role of within-industry firm heterogeneity in migrant share in the welfare effects of a given immigrant shock $\tilde{S}$.

Given that our comparison model is similar to the closest that has been considered in the literature (di Giovanni et al., 2015), the models' comparison also answers the question "How different are the effects of immigration in a model with within-industry firm heterogeneity in immigrant share relative to the closest model of immigration in the literature?"[39]

Table 2 shows that the homogeneous model delivers welfare gains that are 54% of those in the heterogeneous model (0.05pp), due to a larger decline in native-born income and a smaller reduction in the price index.[40] This result aligns with the two mechanisms highlighted in Section 4.2. First, immigration increases the specialization of immigrants and natives in producing different varieties, which weakens the competition faced by natives in the labor market and reduces the downward pressure on their wages. Second, the stronger drop in prices in the heterogeneous model follows from the additional decline in unit costs and being concentrated among larger firms, which have a greater weight in consumers' consumption baskets.

[38] Alternatively, one could compare the effects of the same change in the model primitives (e.g., drop in migration costs). This comparison yields an almost identical quantitative conclusion as in Table 2.

[39] In di Giovanni et al. (2015), firms are heterogeneous in their Hicks neutral productivity but homogeneous in their immigrant share (industries are heterogeneous in their factor shares).

[40] The homogeneous model's welfare effects align with Caliendo et al. (2021), who finds that EU labor market integration raised original members' welfare by 0.04% using a model without firm heterogeneity.

Table 2: Welfare effects with and without firm heterogeneity in the immigrant share

	Open to Trade			Trade Autarky		
	Welfare Workers	Nominal Wage	Price Index	Welfare Workers	Nominal Wage	Price Index
Heterogeneous s_j	**0.10%**	-0.06%	-0.16%	**0.12%**	0.01%	-0.11%
Homogeneous within-sector s_j	**0.05%**	-0.07%	-0.12%	**0.06%**	-0.03%	-0.09%
Homogeneous/Heterogeneous	**54%**					

Notes: For both models, we compute the changes on the key endogenous variables of going from the observed equilibrium to an equilibrium where the number of immigrants is 20% higher. The heterogeneous model is our baseline model, and the homogeneous model does not have within-industry firm heterogeneity in immigrant shares. Columns 1-3 compare the effect for an economy open to trade. Columns 4-6 show the effects for a closed economy.

These results highlight the importance of accounting for the heterogeneity in immigrant-native hiring decisions across firms within sectors. Immigration triggers a reallocation of native workers across firms, with significant aggregate quantitative implications.

6.3 The Quantitative Role of Trade

Motivated by the discussion in section 4.3, this section studies quantitatively whether the gains from immigration are larger or lower when a country is open to trade *due to firm heterogeneity in immigrant shares.* We do so by quantifying the double-difference stated in equation 16 using our fully-fledged quantitative model. We need to know $\tilde{\mathbb{W}}^{heterog,open}$, $\tilde{\mathbb{W}}^{heterog,closed}$, $\tilde{\mathbb{W}}^{homog,open}$ and $\tilde{\mathbb{W}}^{homog,closed}$. Given that we already computed $\tilde{\mathbb{W}}^{heterog,open}$ and $\tilde{\mathbb{W}}^{homog,open}$, we only need to compute the welfare effects in the economy under autarky. To compute $\tilde{\mathbb{W}}^{heterog,closed}$, we proceed in two steps. First, starting from the baseline equilibrium with heterogeneous firms and trade openness, we eliminate international trade by raising the iceberg trade costs high enough. This resulting new equilibrium serves as the initial equilibrium in this economy under trade autarky. Second, we induce the same $\tilde{S}$ as before and compute the real wage change for native-born workers, $\tilde{\mathbb{W}}^{heterog,closed}$. To compute $\tilde{\mathbb{W}}^{homog,closed}$, we follow the same two-step procedure, but now the starting point is the equilibrium with homogeneous firms and trade openness.

We find that international trade dampens the gains from immigration, with most of this dampening driven by the differential importance of firm heterogeneity. In particular, welfare gains in an economy with heterogeneous migrant shares under trade autarky are 0.123%, which is 23% (0.023 pp) larger than the 0.10% gains in the baseline economy

(e.g., $\tilde{\mathbb{W}}^{heterog,closed}/\tilde{\mathbb{W}}^{heterog,open} = 1.23$).

Part of this dampening arises from mechanisms that operate even when firms within a sector employ identical immigrant shares: $\tilde{\mathbb{W}}^{homog,open}$ and $\tilde{\mathbb{W}}^{homog,closed}$ are 0.052% and 0.056% respectively, implying that terms-of-trade effects reduce the gains from immigration by 0.004 pp (approximately 15% of the total dampening effect). This magnitude is remarkably similar to the findings of Caliendo et al. (2021), who study the effects of the EU Enlargement using a representative-firm model, lending credibility to our results. The remaining 85% of the total dampening effects (0.022 pp) is due to the heterogeneity in immigrant shares across firms within sectors, as illustrated in equation 16.

These findings indicate that, given current levels of international trade, firm heterogeneity in migrant shares plays a key role in dampening the gains from immigration. Ignoring this heterogeneity may significantly underestimate the role of international trade in mitigating the gains from immigration.

6.4 Comparing policies that affect firm heterogeneity

Beyond policies that affect the inflow of immigrants into the country, which was studied in prior literature, our model allows us to also study policies that alter the hiring costs of immigrants for specific firms. Appendix Section D.2 studies alternative immigration policies that reduce immigrant hiring costs for a subset of firms. Policies targeting smaller firms in the economy mainly benefit non-tradable sector firms because they face higher hiring frictions.[41] Small firms in the tradable sector are not similarly affected, as the cost reduction does not offset productivity differences with large competitors in the sector. The effects on native workers are positive but small. We find that lowering fixed costs for high-productivity firms in the tradable sector has the largest welfare effects for natives as firms affected face a more elastic demand, and can expand production without crowding out as many natives.

[41] Germany implemented the Competence Center for Securing a Skilled Workforce to help small and medium firms hire immigrants (see here for more details). This policy motivated our counterfactual.

7 Conclusion

This paper documents substantial heterogeneity in immigrant shares across employers, with larger firms employing immigrants more intensively than smaller firms due to fixed costs of hiring immigrants.

Motivated by these facts, we set up a model to study analytically the welfare implication of firm heterogeneity in immigrant shares for the welfare effects of immigration. First, we show how ignoring heterogeneity in migrant shares may bias welfare predictions. Second, we study the implications of firm heterogeneity in immigrant shares for the characteristics of countries that make immigrant flows more valuable. In particular, we show that, due to migrant-share heterogeneity, international trade can either amplify or dampen the welfare gains from immigration.

We calibrate the model to match our novel evidence and use it to assess the quantitative relevance of these theoretical results. We find that a 20% increase in the number of immigrants raises welfare by 0.10% for native-born workers and by 1.23% for firm owners, corresponding to gains of 1.5 billion and 15.3 billion, respectively. Regarding the first analytical result, we find that a model without within-industry heterogeneity in immigrant shares predicts that the welfare gains for native workers would be approximately 50% lower. This means that immigration induces a reallocation of resources across firms with quantitatively important aggregate implications. Regarding the second analytical result, we find that international trade dampens the welfare gains from immigration, with 85% of this dampening effect due to firm heterogeneity in immigrant share. Thus, ignoring the documented immigrant share heterogeneity may significantly underestimate the role of international trade in mitigating the gains from immigration.

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Online Appendix for "Firm Heterogeneity and the Impact of Immigration: Evidence from German Establishments"

Contents of the Online Appendix

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A Empirical Appendix

A.1 Data and Summary Statistics

Our data separates workers into 10 nationality groups: 1) Germany, 2) Europe high-income: France, United Kingdom, Netherlands, Belgium, Austria, Switzerland, Finland, and Sweden, 3) EU middle-income: Italy, Spain, Greece, and Portugal, 4) New member states: countries that joined the EU after 2004, 5) countries of former Yugoslavia not in the EU, 6) Turkey, 7) all other European countries including Russia, 8) Asia-Pacific, 9) Africa and Middle East, and 10) the Americas.

A key variable needed for our analysis is workers' immigration status at a given establishment, but the German social security data records citizenship as opposed to country of birth. Since we are interested in country of birth, we redefine this key variable to make sure we count immigrants properly. The most common recoding is when observing individuals with foreign citizenship who become Germans in the next period. If a worker is recorded as a foreigner for at least two periods, we classify them as an immigrant from the initial citizenship country. A second challenge is that some workers might join the labor market with a foreign citizenship, but they may have grown up in Germany with foreign parents. Our results are robust to recoding workers as natives if they have foreign citizenship and either join the labor force at age 20 or younger without a college degree, or join the labor force at age 25 or younger with a college degree.

For establishment location within Germany, our data includes an administrative subdivision of German states into districts called "Kreis." For part of our analysis, we also group districts into local labor market areas following the analysis of Kropp and Schwengler (2011), who use commuting flows to delineate functional labor markets. Throughout the paper, we also distinguish between tradable and non-tradable sectors. The tradable sector is defined as manufacturing, professional services, and wholesale trade. We complement the German administrative data with publicly available datasets from the World Bank to deflate wages and compute exchange rates, the World Input-Output tables for data on trade and international GDP, and the OECD for aggregate migration data.

In Table A1, we present the average employment, college employment, and immigrant distribution by origin region for our sample. We split the establishments in the sample into the tradable and non-tradable sectors and calculate summary statistics for the years 2003 and 2011.

Table A1: Descriptive Statistics

	Tradable		Non-Tradable	
	2003	2011	2003	2011
N establishments (unweighted)	1,530	1,426	2,148	2,379
Mean Employment	45.0	45.9	39.2	36.5
Mean Employment - College	4.5	5.8	3.0	2.9
Share of employment by origin region				
Germany	90.97%	91.15%	92.66%	91.13%
Europe High-Income	1.03%	0.97%	0.74%	0.70%
EU Middle income	1.94%	1.69%	1.22%	1.40%
EU, joined in 2004	0.63%	0.74%	0.68%	1.22%
Europe, other	0.80%	1.10%	0.73%	1.02%
Turkey	2.73%	2.55%	1.71%	2.06%
Former Yugoslavia	0.79%	0.61%	0.73%	0.70%
Asia - Pacific	0.41%	0.52%	0.76%	0.64%
Africa and Middle East	0.52%	0.46%	0.63%	0.75%
Americas	0.16%	0.21%	0.14%	0.36%

Notes: The sample is restricted to establishments with more than 10 employees.

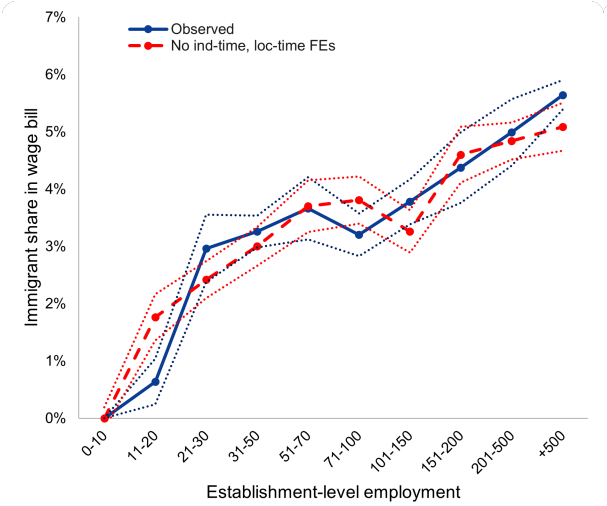
A.2 Additional Facts on Heterogeneity Across Firm Size

We show that our baseline fact cannot solely be explained by large firms having different skill intensities. Large firms tend to be more intensive in high-skill labor (Burstein and Vogel, 2017), and if immigration policy in Germany were skewed toward workers with a specific education group, this could drive the relationship between size and immigrant intensity. However, as shown in Figure A2a, the relationship between size and immigration holds for workers with and without a college education. Moreover, in Figure A2b we also show that our fact holds when excluding multinational companies and when looking at firms that are not part of a multi-establishment firm. The relationship between immigrant share and firm size could also be explained by recent theories on the internal organization of firms, as in Caliendo et al. (2015). If larger firms with more layers of management can supervise and hire more immigrants than smaller firms, it could also rationalize the patterns in Figure 1. However, these theories would not rationalize that larger firms also hire workers from more countries, and expand their immigrant share by increasing the number of source countries, as shown in Figure A6.

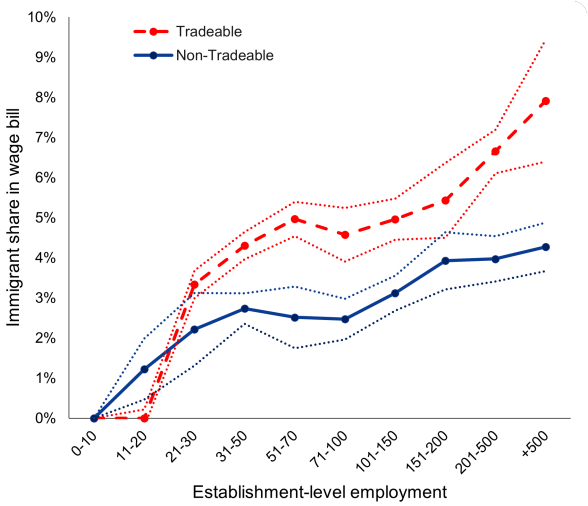
Finally, we corroborate that our findings are not driven just by random matching, where large firms mechanically match with more immigrants. We simulate the hiring of the firms in our sample assuming they draw from a population where 8% are immigrants (based on

Figure A1: Immigrant share across the establishment size distribution

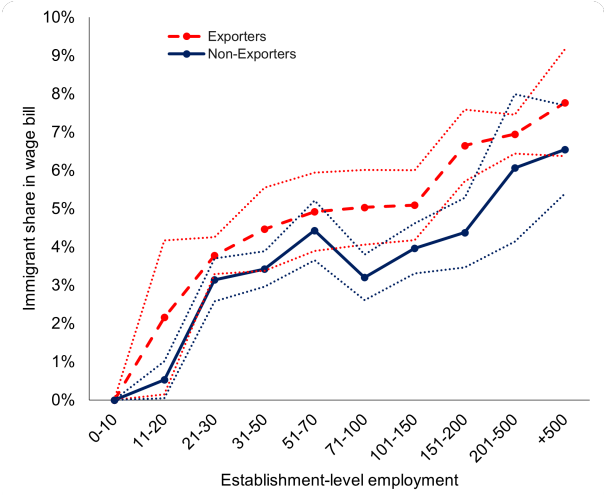
(a) Controlling for industry and geography



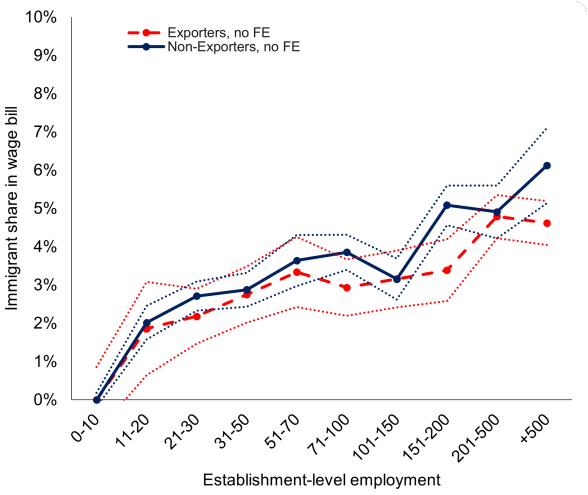
(b) Tradable vs non-tradable sectors



(c) Exporter status



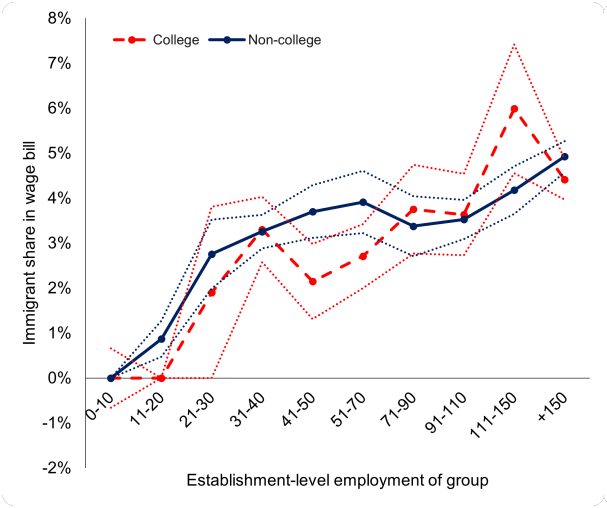
(d) Exporter status, no industry, location



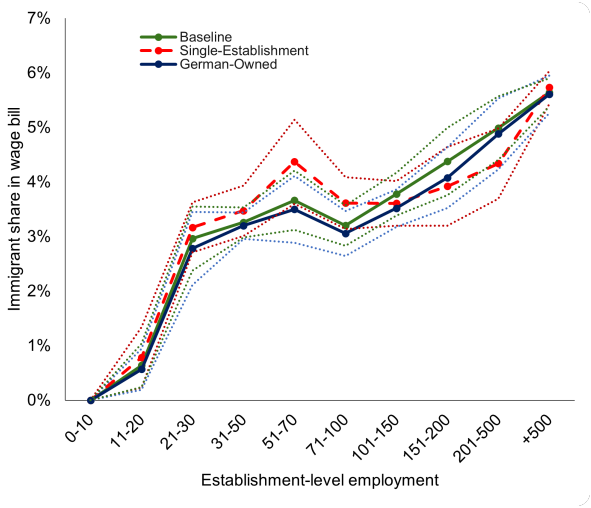
Notes: We divide establishments into bins based on employment size and, for each bin, plot the median immigrant share of the wage bill. We compute 95% confidence intervals using 200 bootstrap repetitions and use the years between 2003 to 2011. To plot Figure A1a, we first regress immigrant shares on 3-digit industry-year and labor market-year fixed effects and plot the median residual for each bin. We then normalize the residual of the smallest bin to 0, only for Figure A1a. In Figure A1b, we plot the median immigrant share for the tradable and non-tradable sectors separately. In Figure A1c we divide the sample between firms that report positive exports that year and firms that do not. In Figure A1d, we do the same split, but first we regress immigrant shares on 3-digit industry and local labor market fixed effects, and plot the residual. We normalize decile 1 to zero for both exporters and non-exporters.

Figure A2: Robustness by skills and establishment type

(a) College vs non-college



(b) Multinationals and multi-unit firms



Notes: We divide establishments into bins based on employment size and, for each bin, plot the median immigrant share of the wage bill. We compute 95% confidence intervals using 200 bootstrap repetitions and use the years between 2003 to 2011. Figure A2a classifies establishments into bins according to their non-college (college) employment and plots for each bin the median share of non-college (college) immigrant wage bill as a share of the non-college (college) workforce. Figure A2b shows the main fact when excluding companies that are foreign-owned (blue line) and companies that are part of a multi-establishment firm (red line).

the aggregate share in the economy). We find that the bin between 0-10 employees has a median immigrant share of zero, but all other bins have a median close to the population average, such that a random matching model would not replicate the positive relationship between immigration and firm size.

A.3 Institutional Background

During our period of analysis, the immigration system in Germany had different requirements depending on the origin of the immigrant workers. At the start of the 21st century, Germany's labor immigration system was largely protectionist, prioritizing EU citizens for job vacancies. Employers who sought to hire non-EU workers had to navigate stringent bureaucratic processes, including the "labor market test", which required proving that no German or EU citizen could fill the position.

Citizens of countries that joined the EU before 2004 (Belgium, France, Italy, Luxembourg, the Netherlands, Denmark, Ireland, the United Kingdom, Greece, Portugal, Spain, Austria, Finland, and Sweden) were free to look for a job without requiring employment sponsorship. During this time period, citizens from other countries were required to have an offer of employment and fulfill certain residence requirements (e.g., evidence of means of subsistence during residence) to receive a permit for employment in Germany. Immigrants also need to obtain consent from the Federal Employment Agency to obtain

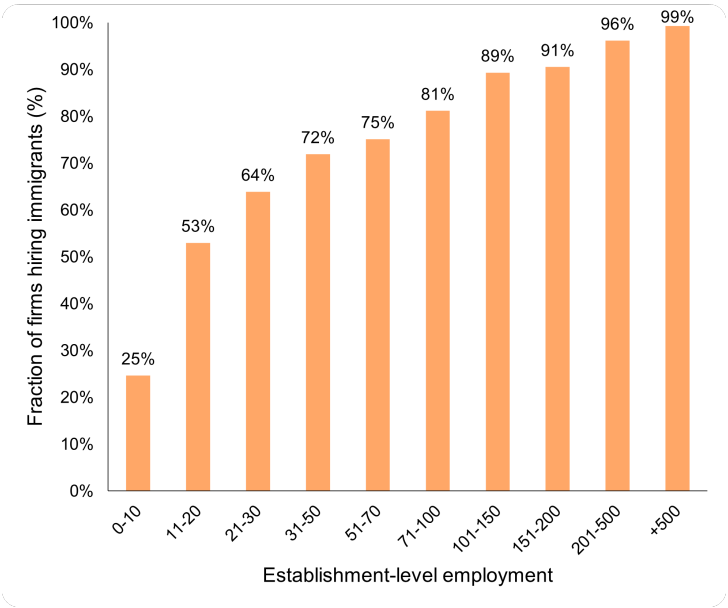
employment.

A.4 Descriptive Evidence of Fixed Costs

This section presents additional stylized facts that motivate our modeling assumptions. We first document evidence consistent with the presence of a fixed cost of hiring immigrants, and then show evidence suggesting that this fixed cost is incurred each time a firm adds a new origin country to its set of immigrant sources.[42]

First, if immigrants and natives are imperfect substitutes, as documented extensively in the literature (Peri and Sparber, 2009, 2011), all establishments would optimally choose to hire a strictly positive level of natives and immigrants in the absence of fixed costs. This contradicts the results in Figure A3, which shows that a significant fraction of establishments do not hire immigrants.[43]

Figure A3: Extensive margin of hiring



Notes: We divide establishments into bins based on employment size. In Figure A3, we plot the fraction of firms in each bin that hire at least one immigrant. We use the years between 2003 to 2011.

Second, we assess whether the fixed costs to hire immigrants may depend on the time spent in Germany by plotting a figure analogous to the motivating figure for long-term

[42] In practice, these fixed costs can be paid either at the firm-level or the establishment-level. For instance, fixed costs such as setting up a legal department and HR protocols might be incurred at the firm level, recruitment-related costs such as learning how to search and screen for suitable candidates are plausible to take place at the establishment level.

[43] While this pattern is consistent with fixed costs, a model where immigrants are randomly matched to establishments could also generate a similar qualitative pattern. However, the size gradient of a random match model would be much steeper than the one observed in the data.

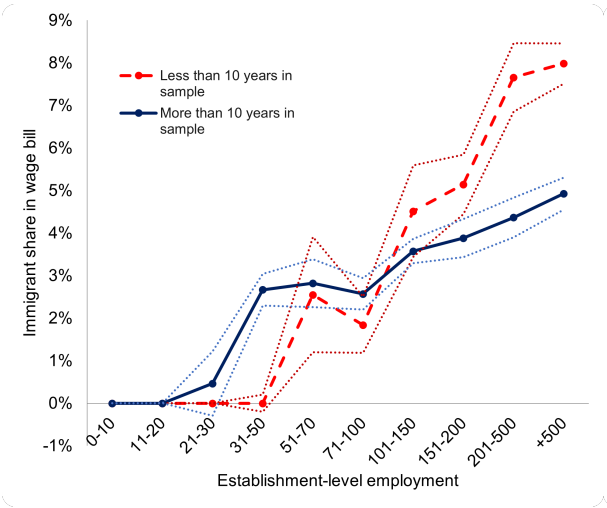
and recent immigrants. We classify workers as long-term or recent immigrants following two common criteria. First, we look at how different our firm heterogeneity looks when comparing immigrants who resided in Germany for over 10 years relative to newer arrivals, inferred from their first appearance in our dataset. We compute the firm-level share of immigrants in a given tenure group as a share of total employees with a similar tenure in our dataset (Figure A4a). Since natives' tenure in the dataset tends to be when they join the labor market while immigrants might have experience before coming to Germany, we look at a second specification where we classify workers into two age groups, whether they are above or below 40 (Figure A4b). This is correlated with time since arrival, as immigrants often arrive in their 20s or early 30s.

Figure A4 shows that the heterogeneity in immigrant share across the firm size distribution is more pronounced among recent immigrants. However, in both cases, the slope for older workers does not flatten completely, indicating that fixed costs associated with hiring immigrants persist relative to hiring natives. This observation is consistent with findings from the existing immigration literature. For example, Peri and Sparber (2009) find that long-term immigrants (over 10 years in the country) perform tasks more similar to natives than recent immigrants, though differences with natives do not fully disappear. Specifically, natives tend to specialize in tasks requiring greater language proficiency compared to long-term immigrants, which German firms have identified as one of the main obstacles to hiring immigrants. There is also persistence in employment, where immigrants might find jobs at big firms when arriving in the country, and then stay at those firms as their career progresses.

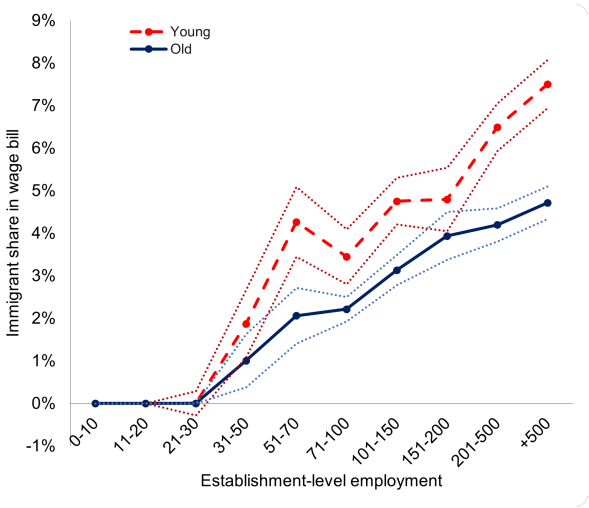
Third, we show that the immigrant hiring process exhibits lumpiness, which is consistent with firms paying a fixed cost. That is, if there are fixed costs to hire immigrants, we should observe a jump in hiring for the initial period followed by a drop toward steady-state levels of hiring thereafter. If there were no fixed costs, firms could begin hiring small amounts and expand their hiring over time. We focus on firms that did not hire immigrants and look at their hiring dynamics after the first period in which they begin hiring. To avoid having our results driven by firms that only hire one immigrant and do not hire again, we restrict the sample to firms that hire at least 5 migrants in our period of study. As these sample restrictions are quite stringent, we use the SIEED dataset because it has a larger sample of firms and a longer time frame.
Figure A5a shows that there is a jump in the number of immigrant new hires, where the average firm hires 3.2 new migrants in the first period and then stabilizes at 2 new migrants every period after that. A similar picture is presented in Figure A5b, when looking at the immigrant share of new hires, which jumps to 40% of hires being migrants and then stabilizes at below 30%.

Figure A4: Differences in heterogeneity across sub-groups

(a) By tenure in the sample



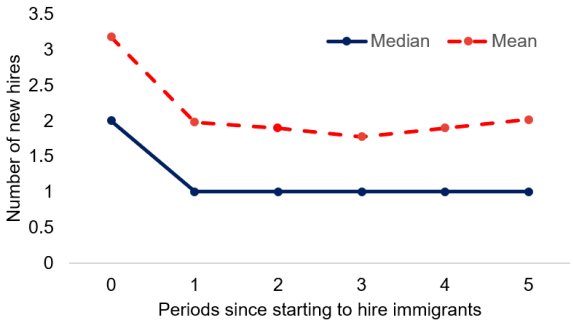
(b) By age



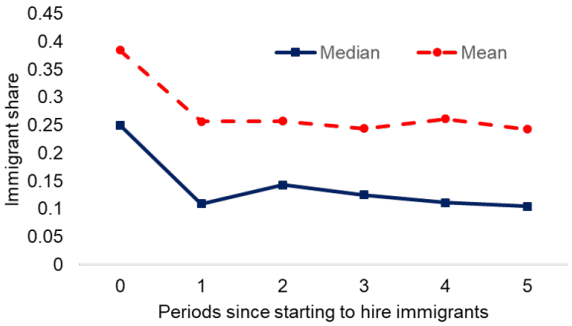
Notes: We divide all establishments into 10 bins according to their reported employment. For each bin, we plot the median immigrant share of the wage bill by sub-group across firms. In Figure A4a compute the immigrant share among workers with more (less) than 10 years of tenure in the sample. In Figure A4b, we compute the immigrant share among workers who are above and below 40 years old. We use the years between 2003 to 2011.

Figure A5: Hiring dynamics after starting to hire immigrants

(a) Number of immigrant new hires



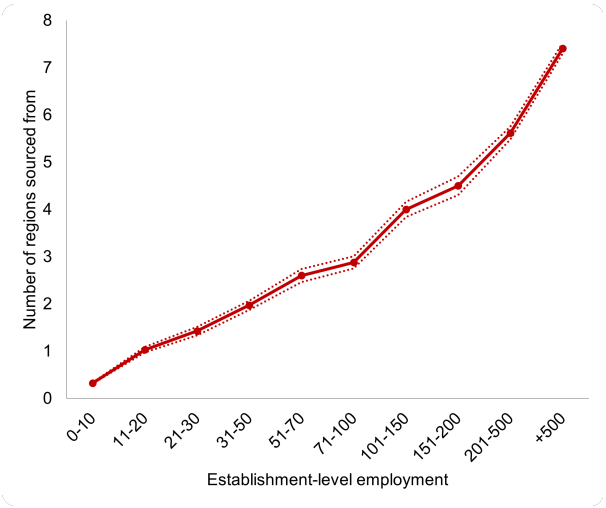
(b) Immigrant share among new hires



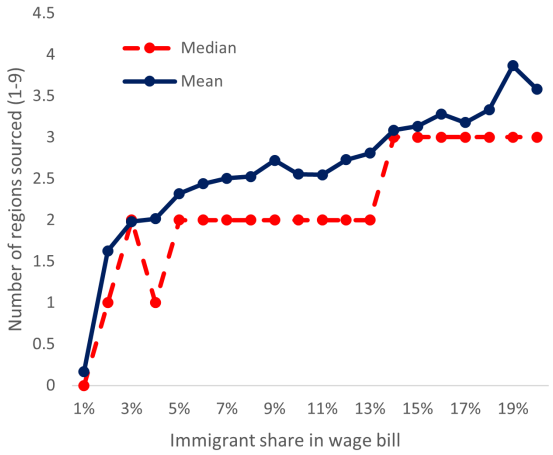
Notes: We restrict the sample to firms that begin to hire immigrants at some point between 2003 and 2010. We then compute the number of new hires that are immigrants each period and the share among new hires that are immigrants. We restrict the sample to those firms that hire at least 5 immigrants during the period. For this specific analysis, we use the SIEED dataset.

Figure A6: Number of regions sourced from

(a) By size



(b) By immigrant share



Notes: In Figure A3, we plot the fraction of firms in each bin that hire at least one immigrant. In Figure A6a, we divide establishments into bins based on employment size and plot for each bin the average number of regions from the rest of the world that the establishment hires workers from. In Figure A6b, we group establishments by the share of the wage bill spent on immigrants into 20 bins (those who spend 0-1%, 1-2%, etc.). For firms in each bin, we plot the mean and median number of origin countries. In our sample, we have 9 immigrant origin regions, which are listed in section 2. We use the years between 2003 to 2011.

Fourth, our data suggest the presence of a fixed cost per source country when hiring immigrants. Appendix Section A.6 shows that immigrants from different origins are imperfect substitutes. In the absence of a fixed cost associated with adding an origin, any firm that hires immigrants would optimally source from all countries in order to exploit these gains from variety. This prediction is inconsistent with Figure A6a, which shows that only the largest establishments hire immigrants from all regions. We therefore model immigrant hiring as involving a fixed cost that must be paid each time a firm expands its set of origin countries. Under this assumption, the model predicts that firms that source immigrants from more countries should be more immigrant-intensive (see equation 9). Consistent with this prediction, Figure A6b shows a strong positive relationship between the number of source countries and the immigrant share of employment. A potential concern is that this relationship may be mechanical, since firms that hire more immigrants may naturally draw from more countries. To address this, Table A2 shows that the correlation between immigrant share and the number of source countries remains strong and statistically significant even after controlling for the total number of immigrants hired. A variance decomposition based on these estimates further indicates that differences along the extensive margin (the number of source countries) account for about 10% of the variation in immigrant shares, compared to only 3% explained by the intensive margin (the number of immigrants hired).

Table A2: Immigrant share: Intensive vs Extensive Margin: OLS estimate

	Immigrant share	Immigrant share
N countries	0.016***	0.012***
	(0.0008)	(0.0009)
N immigrants		5.23e-03
		(1.07e-06)
N observations	17,501	17,501
N establishments	2,485	2,485

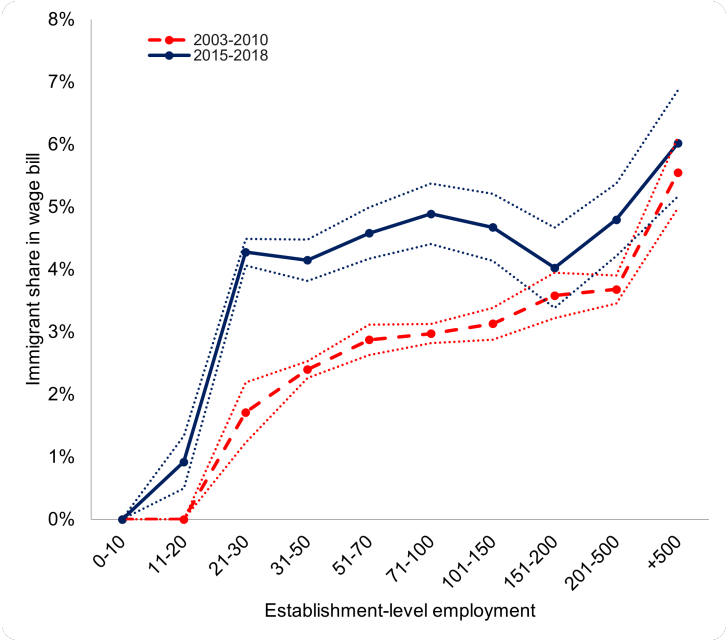
Notes: $***= p < 0.01, ** = p < 0.05, * = p < 0.1$. We control for 2-digit industry-time fixed effects and local labor market time trends. Standard errors are clustered at the establishment level. The sample is restricted to establishments with more than 10 employees.

To conclude, we interpret this evidence as consistent with an environment where large firms are more immigrant-intensive than small firms because they can afford to pay more fixed costs to hire immigrants from different origins.

A.5 Causal Evidence of Fixed Costs to Hire Immigrants

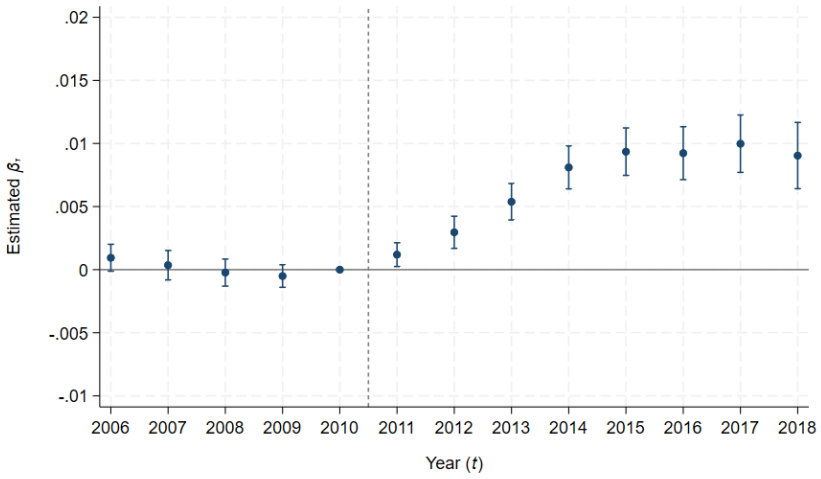
Change in the relationship between immigrant intensity and employer size: We begin by using raw data to show evidence consistent with the policy reducing the fixed cost of hiring NMS immigrants. As explained before, we use the SIEED dataset to analyze the EU enlargement episode. Figure A7 plots the relationship between immigrant share and establishment size for the period before the policy change in red (analogous to Figure 1, which uses our baseline dataset) and for the period after the policy change in blue. Two facts emerge. First, the establishment size threshold above which establishments begin hiring immigrants decreases after the immigration restrictions are lifted, shifting from the 21–30 size bin to the 11–20 size bin. Second, while the median establishment in the top three deciles does not exhibit a significant increase in immigrant share, establishments in other deciles show the largest increases.

Figure A7: Immigrant share for before and after the EU enlargement



Notes: We divide establishments into bins based on employment size and, for each bin, plot the median immigrant share of the wage bill. We compute 95% confidence intervals using 200 bootstrap repetitions. The red and blue lines correspond to the periods 2003-2010 (before the EU enlargement) and 2011-2018 (after the EU enlargement), respectively. To compute this fact, we use the SIEED database (see Section 2 for details).

Figure A8: Evidence of fixed cost to hire immigrants before 2011



Notes: This figure plots the coefficients β_τ from equation 1, without the size triple interaction. The sample is restricted to all establishments that did not hire NMS or other European migrants in the year 2005 (the year before our sample begins). We cluster standard errors at the establishment level and report 95% confidence intervals.

Table A3: Alternative outcomes and size breakdowns

	Employment$_{o,j,t}$	New Hires$_{o,j,t}$	New entrants$_{o,j,t}$	Immigrant share$_{o,j,t}$
	Relative to bottom 3 groups			
$\theta_t \mathbb{1}(o \in NMS)$	0.00627***	0.00301***	0.00161***	0.0317***
	(0.0005)	(0.0003)	(0.0002)	(0.004)
$\theta_t \mathbb{1}(o \in \text{NMS}) \times \mathbb{1}(j \in g = 4)$	-0.00456***	-0.00193***	-0.00135***	-0.0451
	(0.001)	(0.0005)	(0.0003)	(0.028)
	Relative to the largest group			
$\theta_t \mathbb{1}(o \in NMS)$	0.00467***	0.00198***	0.00136***	0.0424
	(0.00106)	(0.000524)	(0.000270)	(0.0275)
$\theta_t \mathbb{1}(o \in \text{NMS}) \times \mathbb{1}(j \in g = 1)$	0.00505***	0.00206***	0.00136***	0.0393
	(0.001)	(0.0005)	(0.0003)	(0.028)
$\theta_t \mathbb{1}(o \in \text{NMS}) \times \mathbb{1}(j \in g = 2)$	0.00275*	0.00135**	0.00144***	0.0489*
	(0.0011)	(0.0005)	(0.0003)	(0.028)
$\theta_t \mathbb{1}(o \in \text{NMS}) \times \mathbb{1}(j \in g = 3)$	0.00154	0.000346	0.000466	0.0683**
	(0.00149)	(0.000690)	(0.000298)	(0.0335)
N	723824	723824	723824	123920

Notes: $***p < 0.01, **p < 0.05, *p < 0.1$, SE are clustered at firm level. In column 1, the outcome is the firm-level employment of workers from o relative to the firm's total employment in 2010. Column 2 shows new hires from o relative to employment in 2010. Column 3 uses the outcome of new entrants to the labor market from origin o relative to firm employment in 2010. Column 4 plots the share of nationality o among all immigrants employed by the firm. The top panel includes the interaction between the treated nationality and the largest size bin (+200 employees), while the bottom panel includes the interactions between the treated nationality and each of the three bottom groups (1-29, 30-99, 100-199 employees, respectively). All specifications include origin-firm and firm-time fixed effects. We restrict the sample to firms that did not hire from treated and control nationalities in 2005.

Table A4: Robustness to alternative samples

	$\text{Employment}_{o,j,t}$	$\text{Employment}_{o,j,t}$	$\text{Employment}_{o,j,t}$	$\text{Employment}_{o,j,t}$
$\theta_t \mathbb{1}(o \in NMS)$	0.00741***	0.00627***	0.00607***	0.00433***
	(0.0006)	(0.0005)	(0.0006)	(0.0006)
$\theta_t \mathbb{1}(o \in \text{NMS}) \times \mathbb{1}(j \in g = 4)$	-0.00304**	-0.00456***	-0.0017	-0.0026**
	(0.0014)	(0.001)	(0.0014)	(0.0011)
Sample	Full sample	Not hiring in 2005	More than 10 employees	Not hiring in 2005 and more than 10 employees
N	769128	723824	171412	141428

Notes: $***p < 0.01, **p < 0.05, *p < 0.1$, SE are clustered at firm level. Column 1 presents the results for the full sample of firms, column 2 (our baseline) presents the results for firms that did not hire from treated and control nationalities in 2005, column 3 restricts to firms that have more than 10 employees in 2010. Column 4 imposes restrictions from columns 2 and 3 simultaneously. All specifications include origin-firm and firm-time fixed effects. The outcome is the employment of nationality o by firm j relative to its total 2010 employment.

Table A5: Robustness to alternative fixed effects

	$\text{Employment}_{o,j,t}$	$\text{Employment}_{o,j,t}$	$\text{Employment}_{o,j,t}$	$\text{Employment}_{o,j,t}$
$\theta_t \mathbb{1}(o \in NMS)$	0.00627***	0.00632***	0.00627***	-
	(0.0005)	(0.0005)	(0.0005)	
$\theta_t \times \mathbb{1}(j \in g = 4)$	-	0.0000323	-0.00342***	-
		(0.0011)	(0.0007)	
$\theta_t \mathbb{1}(o \in \text{NMS}) \times \mathbb{1}(j \in g = 4)$	-0.00456***	-0.00460***	-0.00456***	-0.00502***
	(0.001)	(0.001)	(0.001)	(0.001)
Origin-firm	x	x	x	x
Firm-time	x			x
Origin-time				x
Industry-time market-time		x		
Time			x	
N	723824	720272	723824	723824

Notes: $***p < 0.01, **p < 0.05, *p < 0.1$, SE are clustered at firm level. We restrict the sample to firms that did not hire from treated and control nationalities in 2005. The outcome is the employment of nationality o by firm j relative to its total 2010 employment.

A.6 Differences Between Immigrants and Natives at the Firm

This section provides evidence of the dimensions along which immigrants and natives are different. We begin by using raw data to document that immigrants from different countries tend to specialize in different tasks or occupations than natives and have different "bundles of demographics", such as age, gender, and education. We then show that demographic characteristics and unobservable region-specific characteristics may explain productivity differentials (or comparative advantages) consistent with the observed occupational sorting.

Inspired by the literature started by Roy (1951), we measure sorting into occupations as the share of immigrants from a given country choosing occupation *occ* relative to that share for German workers. This measure is shown in panel A of Table A6. Panel B shows the average age, the share of workers who are females, and the share of workers who have earned a college degree for the same group of countries, relative to Germans.

Table A6 highlights substantial heterogeneity in occupational sorting and demographics across workers from different nationalities. The average German worker is 39 years old, has a 13% probability of holding a college degree, is 41% likely to be female, and is predominantly employed in skilled commercial and administrative occupations. In contrast, sorting patterns among immigrants vary significantly. Turkish workers, for instance, are 13 percentage points less likely to have a college degree, are on average three years younger than Germans, and are concentrated in unskilled manual and service jobs. Immigrants from high-income European countries (France, UK, Netherlands, Belgium, Austria, Switzerland, Finland, Sweden) exhibit the opposite pattern, being 11 percentage points more likely to hold a college degree, three years older than Germans on average,

Table A6: Relative occupational sorting and demographics for immigrants and natives

	All Immigrants	High Income Europe	EU Middle Income	New EU	Turkey	Former Yugoslavia	Europe non EU	Asia	Africa & Middle East	Americas
Panel A										
Agriculture	0.68	0.50	0.58	2.33	0.42	0.60	0.67	0.60	0.54	0.53
Unskilled Manual	1.64	1.09	1.76	1.20	2.02	1.82	1.39	1.54	1.47	0.89
Skilled Manual	0.77	0.58	1.00	0.62	0.93	1.00	0.75	0.46	0.42	0.44
Technicians	0.40	1.09	0.44	0.46	0.24	0.36	0.41	0.40	0.31	0.52
Engineers	0.66	1.78	0.50	0.77	0.23	0.24	0.72	1.44	0.69	1.76
Unskilled Services	1.98	1.02	1.87	1.66	2.18	2.12	1.92	1.79	2.67	1.56
Skilled Services	0.80	1.51	0.65	1.25	0.48	0.67	0.90	0.94	0.78	1.90
Semi-Professions	0.39	0.67	0.29	0.80	0.20	0.32	0.53	0.45	0.36	0.79
Professions	0.93	1.59	0.56	2.11	0.21	0.24	1.32	2.36	0.95	1.99
Unskilled Commercial and Admin	0.84	0.87	0.87	0.88	0.82	0.80	0.96	0.72	0.81	0.87
Skilled Commercial and Admin	0.39	0.84	0.42	0.49	0.26	0.31	0.49	0.39	0.28	0.61
Managers	0.50	1.78	0.39	0.64	0.17	0.19	0.51	0.85	0.32	1.69
Panel B										
Share college	0.77	1.79	0.44	1.42	0.22	0.21	1.02	1.67	0.79	1.88
Share female	0.85	0.80	0.82	1.25	0.70	0.88	1.09	0.97	0.71	0.97
Age	0.97	1.07	1.00	0.99	0.92	1.03	0.95	0.94	0.95	1.00

Notes: Panel A plots the relative share of immigrants from a given origin in a given occupation relative to the share of natives in such occupation. Numbers larger than 1 are interpreted as immigrants being overrepresented in such occupations. Blossfeld occupational classification is used. Panel B plots the relative characteristics of immigrants of a given origin relative to natives. Data for 2003 to 2011.

and concentrated in managerial, engineering, and professional occupations (e.g., doctors, dentists, professors). Middle-income EU immigrants (Portugal, Italy, Spain, Ireland) are overrepresented in unskilled manual and service occupations. Immigrants from new EU 2004 countries are predominantly employed in agricultural and professional occupations, while those from Africa, the Middle East, and Yugoslavia concentrate in unskilled service and manual jobs. Immigrants from Asia tend to be engineers.

To examine occupational differences across nationalities and the determinants of sorting, we follow standard approaches in the immigration literature by distinguishing occupations based on their intensity in manual and communication tasks. Following closely Peri and Sparber (2009) and Peri and Sparber (2011), we assign the manual relative to language task contents to workers based on their occupation, allowing us to quantify the task supply of different nationalities.[44] We then analyze the correlation between the relative task content supplied by workers and their country of origin by estimating equation 22 for worker i from country group o in year t:

$$\left(\frac{M}{L}\right)_{occ(it),t} = FE_{o(i)} + FE_{f(it),t} + \epsilon_{it} \tag{22}$$

where the outcome variable is the ratio of manual to communication task content of worker i's occupation in year t. $FE_{o(i)}$ are country of origin dummies, which are the explanatory variables of interest, and $FE_{f(it),t}$ are employer-year fixed effects, which control for firm-

[44]The US Department of Labor's O*NET abilities survey quantifies the importance of 52 distinct employee abilities, which we refer to as "tasks", for each occupation. Peri and Sparber (2009) group these abilities into categories to construct indices for manual and communication tasks. We map these occupations to German categories using a crosswalk and compute the ratio for each occupation-year.

level shocks and allow us to compare workers within the workplace. We also estimate a version of this regression with an immigrant dummy FE_{imm} rather than $FE_{o(i)}$.

The first column of Table A7 shows that, compared to natives, immigrants are more likely to work in occupations with higher relative manual requirements.[45] On average, the index differs by 0.455, equivalent to 20% of the index for the average occupation. The second column breaks the immigrant group into groups of origin countries. Workers from Turkey are the most specialized in manual-intensive occupations, while workers from high-income European countries specialize in communication-intensive occupations, and workers from the Americas have a similar specialization to Germans. Immigrants from the remaining nationalities tend to specialize in manual-intensive occupations relative to Germans with different degrees of intensity. The third column controls for workers' education, gender, and age. The difference in the estimates between columns 2 and 3 suggests that differences in observable characteristics of workers from different origins explain part of the observed sorting into occupations. For instance, not controlling for worker characteristics more than doubles the specialization for high-income European workers ($-0.156/-0.06 = 2.6$), and increases the specialization of Turkish workers by 20% ($0.765/0.639 = 1.20$).

Next, we provide evidence that the observed occupational sorting may be explained by comparative advantages across origin countries. If sorting is driven by comparative advantages, immigrants' wages relative to natives should be higher in occupations where they have a relative specialization. To test this hypothesis, we estimate equation 23:

$$Log(wage)_{it} = \sum_o \beta_o \, \mathbb{1}(i \in o) \times \left(\frac{M}{L}\right)_{i,t} + \sum_o \alpha_o \, \mathbb{1}(i \in o) + \delta \left(\frac{M}{L}\right)_{i,t} + FE_{f(it),t} + \epsilon_{i,t} \quad (23)$$

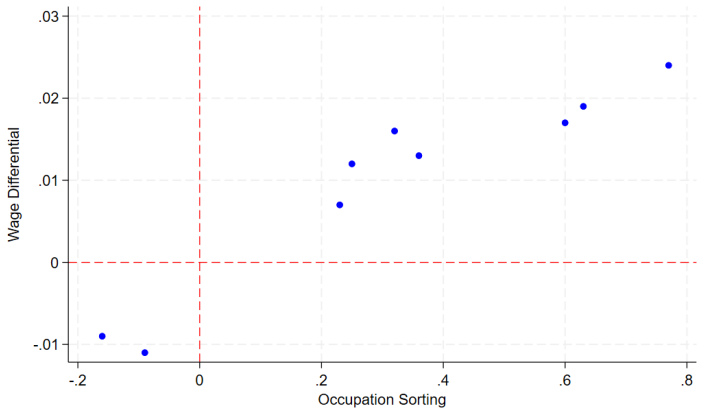
where the omitted category is German nationality. The coefficients of interest are the β's, which capture the wage premium for different immigrant groups relative to Germans in relatively manual-intensive jobs.

The estimates of β are in Columns 4-5 of Table A7. Column 4, which pools all immigrant origins, suggests that there are no significant productivity differences between immigrants and native coworkers along the task distribution. However, this result masks significant heterogeneity in the specialization profiles of workers from different nationalities. Column 5 disaggregates immigrants by nationality and shows that country groups, such as middle-income EU countries, New EU states, Turkey, Yugoslavia, non-EU Europe, Asia, and Africa/Middle East, which tend to sort into manual-intensive tasks, are

[45] This pattern holds within education groups, which is consistent with the findings of Peri and Sparber (2009) for low-skilled workers and Peri and Sparber (2011) for high-skilled workers.

precisely the ones that receive a premium for manual tasks relative to communication tasks. In contrast, country groups that sort into communication-intensive tasks, such as high-income European countries and countries in the Americas, receive a premium for communication tasks relative to manual.[46] In Figure A9 we show the positive relationship between occupational sorting and comparative advantage by plotting the country fixed effects from equation 22 against the wage differentials by country estimated by β_o in equation 22.

Figure A9: Occupational sorting and comparative advantage



Notes: We plot the coefficients for the country fixed effects estimated by equation 22 that capture occupational sorting (horizontal axis) against the β_o coefficients from equation 23, which capture the wage differential when working at a relatively more manual-intensive occupation (vertical axis). We focus on the specification without additional controls.

In summary, workers from different nationalities specialize in different tasks and sort into different occupations accordingly. We provide evidence that productivity differences (or comparative advantage) might be one explanation for this sorting and show what individual characteristics of workers make them useful for specific tasks.

The evidence above is consistent with our modeling assumptions. Specifically, productive sorting into tasks based on comparative advantages is consistent with a CES technology in tasks where efficient labor units from different nationalities are perfect substitutes for a given task and workers from different countries have comparative advantages in specific tasks. As shown by Burstein et al. (2020) and Brinatti et al. (2023), this framework implies the same demand functions as a framework with a CES technology across workers of different nationalities, which is the modeling approach we adopt in this paper. Thus, one interpretation of our CES parameters ϵ and κ is that they capture the elasticity of substitution across nationalities, taking into account that workers of these nationalities have a

[46] We also estimate equation 23 replacing the dummies of nationality by demographic characteristics, and find that younger, non-college-educated, and male workers seem to complement better with manual-intensive jobs relative to older, college-educated, and female workers.

different composition of characteristics, such as different occupations and demographics that make them imperfect substitutes.

As a final point, we look at whether immigrants keep the features that make them different as they spend time working in Germany. Long-term and recent immigrants tend to differ in the types of tasks they perform. Peri and Sparber (2009) find that long-term immigrants (those who have been in the country for over 10 years) engage in tasks with a manual-language composition that is closer to natives than to recent immigrants. However, differences with natives do not fully disappear through the life cycle. To understand whether this pattern holds in our setting, we adopt their approach of comparing the relative manual intensity of occupations as individuals spend more time in Germany. Specifically, we estimate equation 24 separately for each nationality o of immigrants:

$$\left(\frac{M}{L}\right)_{occ(it),t} = \beta_o \, \mathbb{1}(\text{Years in labor market}_i > 10) + X_{i,t} + FE_{f(it),t} + \epsilon_{i,t} \tag{24}$$

where the dummy variable $\mathbb{1}(\text{Years in labor market}_i > 10)$ equals one if worker i has been more than 10 years in the labor market in Germany and zero otherwise. $X_{i,t}$ includes age, education-year fixed effects, and gender-year fixed effects to control for the time-varying factors that may affect the sorting of these workers. $\epsilon_{i,t}$ is the error term, which we cluster at the worker level. The coefficient of interest, β, captures the difference in the manual language intensity of long-term immigrants compared to their compatriots who have been in Germany for less than 10 years.

Table A8 presents the estimates of β for each origin country, showing that long-term immigrants from high-income EU countries and the Americas tend to work in more manual-intensive occupations relative to recent immigrants. These nationalities are precisely those that specialize in linguistically intensive tasks relative to Germans, suggesting that as immigrants spend more time in the country, the task content of their jobs gets closer to that of their native coworkers. Similarly, immigrants from middle-income EU countries, who specialize in manual-intensive tasks relative to Germans, decrease their manual content over time. The differences between long-term and recent immigrants from other nationalities are not statistically significant at a 5% confidence level.

Table A7: Manual to language occupation sorting and comparative advantage

	$\frac{M}{L}_{o(it),t}$	$\frac{M}{L}_{o(it),t}$	$\frac{M}{L}_{o(it),t}$	$log(\text{wage})_{i,t}$	$log(\text{wage})_{i,t}$
$\mathbb{1}(\text{Immigrant} = 1)$	-0.46**			-0.064	
	(0.13)			(0.030)	
$\mathbb{1}(\text{High-income Europe} = 1)$		-0.16***	-0.06*		0.117***
		(0.03)	(0.02)		(0.002)
$\mathbb{1}(\text{Middle-income EU} = 1)$		0.63***	0.55***		-0.078***
		(0.02)	(0.01)		(0.004)
$\mathbb{1}(\text{New Member states} = 1)$		0.23***	0.33***		-0.017*
		(0.03)	(0.03)		(0.006)
$\mathbb{1}(\text{Turkey} = 1)$		0.77***	0.64***		-0.137***
		(0.03)	(0.04)		(0.008)
$\mathbb{1}(\text{Yugoslavia} = 1)$		0.60***	0.53***		-0.080***
		(0.03)	(0.03)		(0.004)
$\mathbb{1}(\text{Europe non-EU} = 1)$		0.32***	0.36***		-0.083***
		(0.03)	(0.03)		(0.004)
$\mathbb{1}(\text{Asia} = 1)$		0.25***	0.34***		-0.089***
		(0.03)	(0.03)		(0.007)
$\mathbb{1}(\text{Africa and Middle East} = 1)$		0.36***	0.29**		-0.090***
		(0.07)	(0.07)		(0.008)
$\mathbb{1}(\text{Americas} = 1)$		-0.09**	0.05		0.023**
		(0.02)	(0.02)		(0.006)
$(M/L)_{o(it),t}$				-0.035***	-0.035***
				(0.001)	(0.001)
$\mathbb{1}(\text{Immigrant} = 1) \times (M/L)_{o(it),t}$				0.013*	
				(0.005)	
$\mathbb{1}(\text{High-income Europe} = 1) \times (M/L)_{o(it),t}$					-0.009***
					(0.001)
$\mathbb{1}(\text{Middle-income EU} = 1) \times (M/L)_{o(it),t}$					0.019***
					(0.001)
$\mathbb{1}(\text{New Member states} = 1) \times (M/L)_{o(it),t}$					0.007***
					(0.001)
$\mathbb{1}(\text{Turkey} = 1) \times (M/L)_{o(it),t}$					0.024***
					(0.002)
$\mathbb{1}(\text{Yugoslavia} = 1) \times (M/L)_{o(it),t}$					0.017***
					(0.001)
$\mathbb{1}(\text{Europe non-EU} = 1) \times (M/L)_{o(it),t}$					0.016***
					(0.001)
$\mathbb{1}(\text{Asia} = 1) \times (M/L)_{o(it),t}$					0.012***
					(0.001)
$\mathbb{1}(\text{Africa and Middle East} = 1) \times (M/L)_{o(it),t}$					0.013***
					(0.002)
$\mathbb{1}(\text{Americas} = 1) \times (M/L)_{o(it),t}$					-0.011***
					(0.002)
Number of observations	1.26E+07	1.26E+07	1.25E+07	1.25E+07	1.25E+07
R^2	0.52	0.52	0.55	0.67	0.67
Controls for worker demographics	N	N	Y	N	N

Notes: $***= p < 0.01, ** = p < 0.05, * = p < 0.1$. Columns 1 -3 run an individual-level regression where the dependent variable is the ratio of manual to language task content for individual i in time t. The explanatory variables are an immigrant dummy (column 1), origin-specific dummies (columns 2 and 3). Column 3 includes controls for age, gender, and education. Columns 4-5, run the regression in equation 23, where the dependent variable is the log wage and the main explanatory variables are country of origin dummies and interactions between country of origin and manual-to-language intensity.

Table A8: Manual to language task intensity by time in Germany

	Germany	Europe high income	EU middle income	EU New member states	Turkey
$\mathbb{1}$(Years in Germany $\geq$ 10)	-0.037***	0.136***	-0.082**	0.075	-0.015
	(0.003)	(0.037)	(0.028)	(0.038)	(0.015)
N	1.12E+07	61,838	121,187	53,014	286,643
	Yugoslavia	Europe non-EU	Asia	Africa and Middle East	Americas
$\mathbb{1}$(Years in Germany $\geq$ 10)	-0.095*	-0.017	0.092*	0.013	0.173**
	(0.042)	(0.028)	(0.038)	(0.021)	(0.066)
N	48,649	68,671	43,399	78,331	11,097

Notes: $***= p < 0.01, ** = p < 0.05, * = p < 0.1$. The outcome of these regressions is the ratio between manual and language task content of individual i's occupation in time t. All regressions include controls for age, education-time fixed effects, and gender-time fixed effects. The explanatory variable is a dummy on whether the individual has been more than 10 years working in Germany. Standard errors clustered at the individual level.

B Theoretical Appendix

B.1 Sourcing Decision

In this section, we derive the immigrant wage index expression in equation 7. Following equation 6, we know the price index for foreign labor is as in the following equation:

$$W_{x,j} = \left(\int_{\Sigma_j} \delta_o^{\kappa} w_{x,k}^{1-\kappa} d_o \right)^{\frac{1}{1-\kappa}} \tag{25}$$

where we use the fact that, in equilibrium, $w_{o,k} = w_{x,k}$ for all $o \in RoW$.[47] δ_o is a source-country specific importance in the production function. We assume δ_o is a Pareto random variable with the following cumulative distribution and density function:

$$F(\delta) = 1 - \left(\frac{\bar{\delta}}{\delta} \right)^{\xi} \quad \text{and} \quad g(\delta) = \xi \bar{\delta}^{\xi} \delta^{-\xi-1} \tag{26}$$

where $\bar{\delta}$ and ξ are the scale and shape parameters, respectively. Since the firm needs to pay a fixed cost f_j for each additional country they hire from, they will just hire from countries with a $\delta > \delta_j^*$, for a given δ_j^*. The mass of countries that the firm hires from is then $n_j = F(\delta > \delta_j^*) = \bar{\delta}^{\xi} (\delta_j^*)^{-\xi}$.

[47] Although we cannot assess the sensitivity of our results to this condition for computational reasons, we conjecture that its quantitative impact is limited because the native workers' welfare depends only on changes in firms' s_j (see equation 40), which the model matches well (validation Figure 3).

With this result, we can calculate the price index of foreign labor as in equation 27:

$$W_{x,j} = \left(w_{x,k}^{1-\kappa} \int_{\delta_j^*}^{\infty} \delta_o^{\kappa} \xi \bar{\delta}^{\xi} \delta^{-\xi-1} d\delta \right)^{\frac{1}{1-\kappa}} = w_{x,k} \left(\left[\frac{\xi \bar{\delta}^{\xi}}{\kappa - \xi} \delta^{\kappa-\xi} \right]_{\delta_j^*}^{\infty} \right)^{\frac{1}{1-\kappa}} = \\ = w_{x,k} \left(\frac{\xi \bar{\delta}^{\xi}}{\xi - \kappa} (\delta_j^*)^{-(\xi-\kappa)} \right)^{\frac{1}{1-\kappa}} \text{ if } \xi - \kappa > 0 \tag{27}$$

Since the mass of countries the firm sources from is $n_j = \bar{\delta}^{\xi}(\delta_j^*)^{-\xi}$, we can now compute the foreign price index as in equation 28:

$$W_{x,j} = w_{x,k} \underbrace{\frac{1}{\bar{\delta}^{\frac{\kappa}{\kappa-1}}} \left(\frac{\xi}{\xi - \kappa} \right)^{\frac{1}{1-\kappa}}}_{\bar{Z}} n^{\underbrace{-\frac{1}{\kappa-1}\frac{\xi-\kappa}{\xi}}_{\iota}} \tag{28}$$

B.2 Equilibrium Equations

The equilibrium in this model is defined as a set of prices, wages, and labor allocations such that: workers optimally choose the industry and destination country d, k to work for, consumers in each location choose how much of each variety to purchase to maximize utility, firms choose the sourcing strategy and export status to maximize profits, labor markets clear, and trade is balanced. Formally, the equilibrium conditions are the following:

1) Consumer budget constraint The total expenditure in Germany (Y_g) and RoW (Y_r) are shown in equation 29:

$$Y_g = \sum_k \left(w_{d,k} L_{d,k} + w_{x,k} L_{x,k} + \Pi_{g,k} \right) \quad Y_r = w_r L_r + \Pi_r \tag{29}$$

where $L_{d,k}$ is the total number of German effective units of labor in sector k, $L_{x,k}$ is the number of effective immigrant units in Germany working in sector k, and $w_{d,k}$, $w_{x,k}$ are the respective effective wages. $\Pi_{g,k}$ are the total profits in sector k in Germany. w_r, L_r, and Π_r are the effective wages, effective labor, and total profits in RoW.

2) Trade balance Total income from exports in Germany is equal to the total import expenditure as in equation 30:

$$\sum_{j \in J_T} \mathbb{1}\left(exporter_j = 1 \right) p_j^x y_j^x = \sum_{j \in J_r} p_j^g y_j^g \tag{30}$$

where J_T and J_r are the sets of firms in Germany in the tradable sector and in RoW, and $p_j^g y_j^g$ are the exports of firm j in RoW to Germany.

3) Labor market clearing The market clearing conditions 31-33 require that demand for effective units of native and immigrant labor equals supply in each industry and country:

$$\sum_{j\in J_k}\left(d_j + n_j f_j + f_k^{imm}\mathbb{1}(n_j>0) + \mathbb{1}\left(exporter_j = 1\right) f_x\right) = \Gamma\left(1-\frac{1}{\nu}\right) A_{g,k,g}^{\frac{1}{\nu}} \left(\pi_{g,k,g}\right)^{\frac{\nu-1}{\nu}} N_g \tag{31}$$

$$\sum_{j\in J_k} x_{j,k} = \Gamma\left(1-\frac{1}{\nu}\right) A_{x,k,g}^{\frac{1}{\nu}} \left(\pi_{x,k,g}\right)^{\frac{\nu-1}{\nu}} N_r \tag{32}$$

$$\sum_{j\in J_r} d_j = \Gamma\left(1-\frac{1}{\nu}\right) A_{x,r}^{\frac{1}{\nu}} \left(\pi_{x,r}\right)^{\frac{\nu-1}{\nu}} N_r \tag{33}$$

where N_r is the total number of workers in RoW and equation 32 uses the fact that $\pi_{o,k,g} = \pi_{x,k,g}$.

B.3 Derivation of Analytic Results: Welfare Effects

We consider a simplified version of our fully quantitative model, where the economy has only one sector and all firms hire some amount of immigrants (e.g., $f_k^{imm} = 0$). We first study the role of firm heterogeneity in immigrant share when the economy is in trade autarky, and later analyze how the results change when the economy is open to trade. We derive the expression for the change in the welfare of native workers in four steps.

Step 1: Express $\tilde{s}_j$ as proportional to $\tilde{s}_1$. The profit function and the corresponding first order condition with respect to s_j are:

$$\begin{aligned}\Pi_j &= A\psi_j^{\sigma-1}s_j^{-\chi} - Bf_j(s_j^{-1}-1)^{\theta+1}\\ \psi_j^{\sigma-1}s_j^{-\chi+1+\theta} &= f_j C(1-s_j)^{\theta}\end{aligned}$$

where $A, B,$ and C are general equilibrium variables that are common to all firms, $\chi \equiv \frac{\sigma-1}{\epsilon-1} > 0$ and $\theta \equiv \left(\iota(\epsilon-1)\right)^{-1} - 1 > 0$.

The first order condition for firm j and firm 1 implies that:[48]

$$\left(-\chi + 1 + \frac{\theta}{1-s_j}\right)\tilde{s}_j = \left(-\chi+1+\frac{\theta}{1-s_1}\right)\tilde{s}_1$$

[48] where $\alpha_j > 0$ if and only if $\epsilon - \sigma + \iota(1-s_j) > 0$.

or

$$\tilde{s}_j = \frac{\alpha_j}{\alpha_1}\,\tilde{s}_1 \quad with \quad \alpha_j = \frac{1}{-\chi + 1 + \theta(1-s_j)^{-1}} \tag{34}$$

Step 2: Express $\tilde{s}_j$ as proportional to $\tilde{S}$.

Let WB_{dj} and WB_j be the native and total wage bill of $j \in J_d$. Let S be the associated aggregate native share:

$$S \equiv \frac{\sum\limits_{j\in J} WB_{dj}}{\sum\limits_{j\in J} WB_j} = \sum_{j\in J} \underbrace{\frac{WB_j}{\sum\limits_{j\in J} WB_j}}_{\equiv\, \omega_j^{wb}} s_j = \sum_{j\in J} \omega_j^{wb} s_j$$

Then the change in the aggregate native share is given by:

$$\tilde{S} = \sum_{j\in J} \underbrace{\frac{\omega_j^{wb} s_j}{\sum\limits_{j\in J} \omega_j^{wb} s_j}}_{\omega_j^S} \left(\tilde{\omega}_j^{wb} + \tilde{s}_j\right) \tag{35}$$

where ω_j^S is the share of firm j in S.[49]

Next, we find an expression for $\tilde{\omega}_j^{wb}$ as a function of $\tilde{s}_j$. To that end, we use firm j's optimal demand for natives and the definition of ω_j^{wb}:

$$WB_j = \frac{\sigma-1}{\sigma} r_j = \frac{D}{\psi_j^{1-\sigma}} s_j^{-\chi} \quad \rightarrow \quad \tilde{WB}_j = \tilde{D} - \chi \tilde{s}_j$$

$$\omega_j^{wb} = \frac{WB_j}{\sum\limits_{l\in J_d} WB_l} \quad \rightarrow \quad \tilde{\omega}_j^{wb} = \tilde{WB}_j - \sum_{l\in J_d} \omega_l^{wb} \tilde{WB}_l$$

where D is a general equilibrium variable common to all firms.
Combining these last two expressions yield $\tilde{\omega}_j^{wb}$ as a function of $\tilde{s}_j$:

$$\tilde{\omega}_j^{wb} = -\chi \left(\tilde{s}_j - \sum_{l\in J_d} \omega_l^{wb}\, \tilde{s}_l\right) \tag{36}$$

This expression, together with 34 and 35, implies that the change in aggregate share can

[49] For simplicity, we define these shares to exclude the employment related to f_j.

be expressed as a function of the change in s_1:

$$\begin{aligned}\tilde{S} &= \sum_{j\in J}\omega_j^S\Big(-\chi\Big(\tilde{s}_j - \sum_{l\in J_d}\omega_l^{wb}\,\tilde{s}_l\Big) + \tilde{s}_j\Big)\\ \tilde{S} &= \sum_{j\in J}\omega_j^S\Big(-\chi(\alpha_j - \sum_{l\in J_d}\omega_l^{wb}\alpha_l) + \alpha_j\Big)\frac{\tilde{s}_1}{\alpha_1}\end{aligned} \tag{37}$$

In a more compact way, it reads as:

$$\tilde{S} = \sum_{j\in J}\omega_j^S\underbrace{\Big(-\chi(\alpha_j - \bar{\alpha}) + \alpha_j\Big)}_{\beta_j}\frac{\tilde{s}_1}{\alpha_1} \tag{38}$$

with $\bar{\alpha} \equiv \sum_{l\in J_d}\omega_l^{wb}\alpha_l$.[50]

We can use equations 34 and 38 to express $\tilde{s}_j$ as proportional function of $\tilde{S}$:

$$\tilde{s}_j = \frac{\alpha_j}{\beta}\tilde{S} \quad with \quad \beta = \sum_{l\in J_d}\omega_l^S\beta_l \tag{39}$$

Step 3: Express welfare change into a component observable with aggregate data and a component that requires micro-level data.

The welfare gains from immigration in this simplified model are given by the drop in the price index induced by immigration. The change in the price index is a weighted average of the changes of individual prices which, in turn, are proportional to the change in the native share:

$$\begin{aligned}\tilde{P} &= \sum_{j\in J}\lambda_j\tilde{p}_j\\ &= \sum_{j\in J}\lambda_j\tilde{u}_j\\ &= \sum_{j\in J}\lambda_j\Big(\tilde{w}_d + \frac{\tilde{s}_j}{\epsilon-1}\Big)\\ &= \tilde{w}_d + \frac{\sum_{j\in J}\lambda_j\tilde{s}_j}{\epsilon-1}\end{aligned} \tag{40}$$

where λ_j is firm j's share in the expenditure of consumers, which coincides with the market share of the firm denoted by ω_j, $\lambda_j = \frac{p_j^{1-\sigma}}{P^{1-\sigma}} = \frac{p_j q_j}{\int p_j q_j} \equiv \omega_j$. The second equality uses the optimal pricing condition, the third equality uses 5, and the fourth equality uses

[50] If all firms choose the same immigrant-share, $\tilde{S} = \tilde{s}_j$.

$\sum_{j \in J} \lambda_j = 1$.

Next we solve for the real wage change and substitute $\tilde{s}_j$ using equation 38 to obtain:

$$\begin{aligned}
\tilde{w}_d - \tilde{P} &= -\frac{\tilde{S}}{\epsilon - 1} \sum_{j \in J} \omega_j \frac{\alpha_j}{\beta} \\
&= -\frac{\tilde{S}}{\epsilon - 1} \left(\frac{\sum_{j \in J} \omega_j^S \beta_j}{\sum_{j \in J} \omega_j \alpha_j} \right)^{-1} \\
&= -\frac{\tilde{S}}{\epsilon - 1} \left(\frac{\sum_{j \in J} \omega_J^S \Big(-\chi(\alpha_j - \bar{\alpha}) + \alpha_j \Big)}{\bar{\alpha}} \right)^{-1} \\
&= -\frac{\tilde{S}}{\epsilon - 1} \left(1 - \frac{\sigma - \epsilon}{\epsilon - 1} \frac{\sum_{j \in J} \omega_j^S \alpha_j - \sum_{j \in J} \omega_j \alpha_j}{\sum_{j \in J} \omega_j \alpha_j} \right)^{-1} \\
&= -\frac{\tilde{S}}{\epsilon - 1} \left(1 - \frac{\epsilon - \sigma}{\epsilon - 1} \pi \right)^{-1}
\end{aligned} \tag{41}$$

where $\pi \equiv 1 - \frac{\sum_{j \in J} \omega_j^S \alpha_j}{\sum_{j \in J} \omega_j \alpha_j}$, which equals zero if firms employ the same immigrant share, and positive otherwise. The first equality uses that $\lambda_j = \omega_j$, the third and fourth equality use that the firm's share in the wage bill coincides the share in shares, $\omega_j^{wb} = \omega_j$.[51]

Next, we show that $\pi \in [0, 1)$, which implies the sign of the bias is determined by the sign of $\epsilon - \sigma$.

Step 4: Determine if the bias is larger or smaller than one.

There is a tight relationship between ω_j and ω_j^S. Specifically:

$$\omega_j^S = \omega_j \frac{s_j}{\sum_{j \in J} \omega_j s_j}$$

This equation implies that the weighting system ω^s assigns a lower weight to immigrant-intensive firms than the weighting system ω. Given that α_j is strictly increasing in the immigrant-share of the firm, the average of α_j under the weighting system ω^s must be

[51] $WB_j = w_d d_j + \sum_c w_x x_{cj} = w_d d_j + w_{jx} x_j = \frac{u_j}{\psi_j} y_j = \Big(\frac{\sigma - 1}{\sigma} \, p_j \Big) y_j = \frac{\sigma - 1}{\sigma} \, p_j y_j$

lower than that under ω_j. Therefore:

$$1-\pi = \frac{\sum_{j\in J} \omega_j^S \alpha_j}{\sum_{j\in J} \omega_j \alpha_j} \in (0,1]$$

Given that this term is always negative, the bias will be higher or lower than one, depending on whether ϵ is larger than σ. If $\epsilon > \sigma$, the last bracket in 41 is lower than one and vice versa.

Economy open to trade:

We now derive the welfare gains in an open economy where trade is balanced, $\sigma = \sigma_x$, and firms must pay a fixed cost to export and iceberg trade costs τ.

The steps of the proof are analogous to the closed economy, but the price index in step 3 now should take into account that imported goods may become more expensive as immigrants reallocate from RoW to Germany. Specifically,

$$\begin{aligned}
\tilde{P} &= \sum_{j\in J} \lambda_j \tilde{p}_j \\
&= \sum_{j\in J} \lambda_j \tilde{u}_j \\
&= \sum_{j\in J_d} \lambda_j \left(\tilde{w}_d + \frac{\tilde{s}_j}{\epsilon - 1}\right) + \sum_{j\in J_x} \lambda_j \tilde{w}_r \\
&= \tilde{w}_d + \frac{\sum_{j\in J_d} \lambda_j \tilde{s}_j}{\epsilon - 1} + (1-\lambda)(\tilde{w}_r - \tilde{w}_d)
\end{aligned} \tag{42}$$

where J_d and J_x are the sets of domestic and foreign varieties or firms, respectively; and λ is Germany's domestic trade share $\lambda \equiv \sum_{j\in J_d} \lambda_j = 1 - \sum_{j\in J_x} \lambda_j$.

To relate this expression to those in the closed economy case, it will be helpful to write the expenditure share λ_j in equation 42 in terms of the domestic sales share: $\lambda_j = \lambda \omega_j$

By following analogous steps as in 41, we can rewrite the welfare effects as follows:

$$\begin{aligned}
\tilde{w}_d - \tilde{P} &= -\lambda \frac{\tilde{S}}{\epsilon - 1}\left(1 + \frac{\epsilon - \sigma}{\epsilon - 1}\pi\right)^{-1} - (1-\lambda)(\tilde{w}_r - \tilde{w}_d) \\
&= -\lambda \frac{\tilde{S}}{(1-\pi)\epsilon + \pi\sigma - 1} - (1-\lambda)\left(\tilde{w}_r - \tilde{w}_d\right)
\end{aligned} \tag{43}$$

B.4 Model with Heterogeneity in Home-Bias Technology $\beta_{k,j}$

This section compares our fixed-cost model with a model featuring heterogeneous home-biased technology $\beta_{k,j}$ (henceforth, the "home-bias model"). Although the home-bias model can also match the cross-sectional variation in immigrant shares, it implies the same elasticity of substitution between immigrants and natives across firms, which is counterfactual.[52] For this reason, we adopt the fixed-cost model and explain how its predictions differ from those of the home-bias model.

The equations analog to 13 and 14 for the home-bias model are:

$$\begin{aligned} &\text{Elasticity of unit cost:} && \tilde{u}_j = (1 - s_j)\, \tilde{w}_x \\ &\text{Elasticity of substitution:} && \widetilde{(1 - s_j)} - \tilde{s}_j = -(\epsilon - 1)\tilde{w}_x \end{aligned}$$

These differences with respect to equations 13 and 14 have relevant aggregate implications. First, in the home-bias model, immigration reduces the immigrant market wage and, all else equal, the marginal cost of production drops proportionally to the share of immigrants in the cost of production. This elasticity of marginal cost of production is stronger in the model with fixed costs because the drop in costs increases profits, and so the firm can pay more times the fixed cost to source immigrants from more origins, which further reduces the unit cost of production. Also, in the home-bias model, the elasticity of substitution between immigrants and natives is the same across firms $(\epsilon - 1)$. In a model with fixed costs, on the contrary, the elasticity of immigrant-to-native share is different across firms. These two differences matter in understanding how and how much resources are reallocated across firms and, therefore, how the goods and labor markets adjust to the inflow.

Second, the aggregate elasticity of substitution between immigrants and natives is enough to quantify the effect of immigration on native welfare in the home-bias but it is not enough in the model with fixed costs. In particular, equation 12 for the the home-bias model is:

$$\tilde{\mathbb{W}} = -\frac{\tilde{S}}{(1 - \pi)\,\epsilon + \pi\,\sigma - 1} = -\frac{\tilde{S}}{\epsilon^{agg} - 1}$$

where $\pi = \pi(\psi_j, \beta_{k,j})$.[53] Thus, if the researcher believes that the heterogeneity in s_j is due to heterogeneous $\beta_{k,j}$, she could quantify the welfare effects (to a first-order approximation) of a given inflow by using an estimate of the aggregate elasticity from the literature, which is available for a wide range of settings. However, if the true model is with fixed costs, she will need the micro elasticities and firms' shares for welfare analysis.

[52] Firms' FOC determine differences in s_j within industry according to $\frac{s_j}{1-s_j} = \frac{\beta_{k,j}}{1-\beta_{k,j}} \left(\frac{w_{x,k}}{w_{d,k}}\right)^{\epsilon-1}$.

[53] More generally, equation 15 holds where $\pi = \pi(\psi_j, \beta_j)$.

In addition to the different aggregate implications, the two models have different policy implications. The fixed cost model has the relative advantage of giving a specific policy tool that can be used to change the immigrant intensity of firms. There were policies in Germany that aimed to ease the frictions of specific firm groups to hire immigrants. Such policies are more reasonable in a context where firms face hiring costs than if the heterogeneity stems from innate differences in a firm's production function.[54]

C Estimation and Validation Appendix

C.1 Estimation of ϵ

To estimate the elasticity of substitution between immigrants and natives, we use the first-order conditions with respect to d_j and x_j to obtain equation 44:

$$Log\left(\frac{\text{Wage Bill Immig}_{j,t}}{\text{Wage Bill Natives}_{j,t}}\right) = \frac{\epsilon-1}{\epsilon} Log\left(\frac{x_{j,t}}{d_{j,t}}\right) + Log\left(\frac{1-\beta_k}{\beta_k}\right) \tag{44}$$

where we add a year subindex t to endogenous variables. This equation shows that ϵ regulates the change in firms' labor spending in native workers relative to immigrants to changes in their relative efficient units of labor.

Given that the efficient units of native labor d_j and the CES composite of immigrant labor x_j are not directly observable, we proceed in two steps to take this equation to the data. First, given that the ability draws η are Fréchet distributed we can write the effective units of domestic labor, $d_{j,t} = \gamma_{g,k,t}\ N_{g,j,t}$ where $\gamma_{g,k,t}$ is the average ability of native workers in industry k and $N_{g,j,t}$ is the number of native workers in firm j in year t.

Second, we need to estimate the quantity of immigrant labor employed by company j, which is given by the following CES composite:

$$x_{j,t} = \left(\sum_o \delta_{o,k} x_{j,o,t}^{\frac{\kappa-1}{\kappa}}\right)^{\frac{\kappa}{\kappa-1}} = \left(\sum_o \delta_{o,k} (\gamma_{o,k,t} N_{o,j,t})^{\frac{\kappa-1}{\kappa}}\right)^{\frac{\kappa}{\kappa-1}} = \left(\sum_o \zeta_{o,k,t} \left(N_{o,j,t}\right)^{\frac{\kappa-1}{\kappa}}\right)^{\frac{\kappa}{\kappa-1}} \tag{45}$$

where we added year sub-indexes t. $\gamma_{o,k,t} \equiv A_{o,k}^{\frac{1}{\nu}} \left(\pi_{o,k,t}\right)^{-\frac{1}{\nu}} \bar{H}$ is the average ability of nationals from o in sector k in year t and the second equality follows from assuming that $A_{o,k}$ is Fréchet distributed.

[54] Motivated by these policies, Section 6.4 shows alternative policies can have significantly different welfare effects.

To compute $x_{j,t}$, we need to estimate the two unobservable components of the right-hand side: the elasticity of substitution κ and the productivity shifters $\zeta_{o,k,t}$. We do so by following Ottaviano and Peri (2012)'s methodology. First, we use the first-order conditions for $x_{o,j,t}$ relative to $x_{o',j,t}$ to derive the following estimating equation 46:

$$\begin{aligned} dLog\left(\text{Wage Bill}_{o,j,t}\right) = \underbrace{\frac{\kappa-1}{\kappa}}_{\beta} dLog(N_{o,j,t}) + \underbrace{dLog(\delta_{o,k,t}) + \frac{\kappa-1}{\kappa} dLog(\gamma_{o,k,t})}_{\zeta_{o,k,t} \text{ Origin-Industry-year FE}} \\ \underbrace{- dLog(\delta_{o',k,t}) - \frac{\kappa-1}{\kappa} dLog(\gamma_{o',k,t} N_{j,t}^{o'})}_{\text{Firm-year FE}} \end{aligned} \tag{46}$$

Second, we use the OLS estimates of β and the origin country-industry-year fixed effects to infer κ and $\zeta_{o,k,t}$ respectively. Finally, we compute $\hat{x}_{j,t}$ by plugging these estimates in equation 45.

In practice, we estimate the regression in levels (with the corresponding firm-origin fixed effect) rather than in changes to increase the precision of the estimates, i.e.,

$$Log\left(\text{Wage bill}_{o,j,t}\right) = \beta\ Log(N_{o,j,t}) \ + \ FE_{o,j} \ + \ FE_{o,k,t} \ + \ FE_{j,t} \ + \ u_{o,j,t} \tag{47}$$

where $u_{o,j,t}$ is a residual and standard errors are clustered at the origin-firm level. Our estimate of β is 0.95 (CI between 0.93 and 0.97), suggesting a value for $\hat{\kappa} = 20.2$ (CI between 12.6 and 27.6). As mentioned earlier, finding comparable estimates is challenging due to the limited availability of firm-level data on employment by nationality. Keeping this limitation in mind, we find that our estimate is consistent with Busch et al. (2020) who calibrate an aggregate elasticity of substitution between immigrants from different nationality groups (OECD vs. non-OECD) in Germany at 22.6.

These estimates of κ and $\zeta_{o,k,t}$ allow us to estimate $x_{j,t}$ and replace the unobservable $d_{j,t}$ and $x_{j,t}$ by functions of observables $N_{d,j,t}$ and $\hat{x}_{j,t}$ respectively, and rewrite 44 as:

$$Log\left(\frac{\text{Wage Bill Immig}_{j,t}}{\text{Wage Bill Natives}_{j,t}}\right) = \frac{\epsilon-1}{\epsilon} Log\left(\frac{\hat{x}_{j,t}}{N_{g,j,t}}\right) - Log\Big(\gamma_{g,k,t}\Big) + Log\left(\frac{1-\beta_k}{\beta_k}\right) + \xi_{j,t} \tag{48}$$

where $\xi_{j,t} \equiv Log(x_{j,t}) - Log(\hat{x}_{j,t})$ is a measurement error term.

To take equation 48 to the data, we set up equation 18, where the industry-time fixed effects should control for $Log\Big(\gamma_{g,k,t}\Big) + Log\left(\frac{1-\beta_k}{\beta_k}\right)$.

Table C1 presents the OLS and 2SLS estimates of β from equation 18 and ϵ. Our OLS estimate for β is 0.88 with a 95% confidence interval (CI) between 0.85 and 0.92. The similarity between the OLS and 2SLS estimates suggests that the stringent set of

fixed-effects may account for a large fraction of the endogeneity concerns, and that firm-level shocks biased toward immigrants relative to natives may not confound the effect of interest.

Table C1: Estimates for ϵ

Method	Parameter	Value	95% Conf. Interval
OLS	$(\epsilon-1)/\epsilon$	0.88***	[0.85 , 0.92]
	Implied ϵ	8.49***	[6.69 - 11.89]
2SLS	Estimate for $(\epsilon-1)/\epsilon$	0.87***	[0.78 , 0.97]
	Implied ϵ	7.95***	[4.33 , 29.58]

Notes: $^{***} = p < 0.01, ^{**} = p < 0.05, ^{*} = p < 0.1$. This table reports the OLS and 2SLS estimation output for equation 48. Confidence intervals are bootstrapped with 5,000 repetitions as described in Section 5.2. The number of observations is 4382. The first-stage F stat is 78.74.

C.2 Empirical Distribution for the Statistic $\widehat{\epsilon-\sigma}$

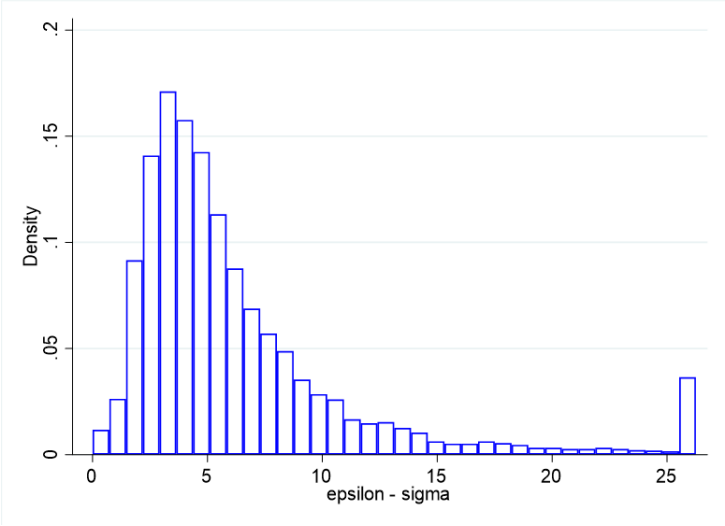
We obtain an empirical distribution for the statistic $\widehat{\epsilon-\sigma}$ to test the null hypothesis that $\epsilon - \sigma < 0$. We use a bootstrap procedure to obtain these empirical distributions that takes into account the correlation between $\hat{\sigma}$ and $\hat{\epsilon}$. Using 5,000 bootstrapped samples, we re-estimate ϵ using equation 18 and σ using equation 17. To calibrate σ under the assumption imposed in Section 4.2, namely $\sigma = \sigma_d = \sigma_x$, we pool firms from both the tradable and non-tradable sectors to compute the average markup. We find that $\widehat{\epsilon-\sigma} < 0$ in only 0.007% of the 5000 cases. Figure C1 plots the distributions of $\widehat{\epsilon-\sigma}$.

C.3 Tests of the Identification Strategy

Our instrument exploits differential exposure of firms to immigrant arrivals from different countries, where the firm exposure is based on the nationality composition of their workforce. Thus, our setting relies on an identifying assumption in terms of the shares Goldsmith-Pinkham et al. (2020). This section describes the variation underlying our instrument and tests the identifying assumption.

Shift-share Instrument Diagnostics: We describe the variation that the instrument uses by calculating the Rottemberg weights of the Bartik instrument. To do this, we write the first stage coefficient on the shift-share instrument as a combination of the estimates of nine separate first stage regressions. Each of these "just identified" regressions uses an instrument that is constructed with the initial share and shock of only one of our nine origin regions. The weights in which each of these nine instruments affects the overall instrument are called Rottemberg weights, which we calculate using the code provided by Goldsmith-Pinkham et al. (2020) and denote as α. Each origin region is affected each

Figure C1: Distribution of $\widehat{\epsilon - \sigma}$



Notes: The distribution includes the realization of $\hat{\epsilon} - \hat{\sigma}$ from 5,000 bootstrapped samples, were we estimate ϵ using equation 48 and σ using equation 17. The fraction of observations for which $\widehat{\epsilon - \sigma} < 0$ is 0.007, rejecting the null hypothesis that $\epsilon - \sigma < 0$ at a 1% confidence. For clearer visualization of the distribution, we replaced negative values with zero and the top 2.5% values with the 97.5th percentile value.

year by a national-level shock, which we denote by G. The just-identified coefficients are denoted by β.

As shown in Table C2, panel A, all the Rottemberg weights are positive, meaning that our regression is likely not subject to misspecification and our coefficient is likely to have a local average treatment effect-like interpretation. Panel B shows the correlation between the weights, the shocks, and the just-identified coefficients. Panel C shows the top five origin regions in terms of the Rottemberg weights, which indicates they are likely to play an important role in the variation of our instrument. The country with the largest weight is middle-income EU countries (0.175), followed by New EU countries (0.143), other European countries not in the EU (0.143), high-income EU countries (0.136), and Turkey (0.125). It is reassuring that no single region accounts for a large majority of the variation in our instrument.

Pre-trend analysis: We assess the plausibility of the identifying assumption with pre-trend tests by estimating the same regressions as in 48 with lagged outcome variables. Failing to reject that an inflow in t had an effect on lagged outcomes would increase the plausibility of the assumption that the common shock caused the change in the changes. Table C3 shows that the effects in the years leading up to the inflow, including up to a five-year lag, are generally insignificant. The only exception is the period immediately preceding the inflow, where the effect is marginally significant (e.g., the lower

Table C2: Shift-share diagnostics

Panel A	Sum	Mean	Share
$\alpha_s \leq 0$	-	-	0
$\alpha_s > 0$	0.99	0.11	1
Panel B	α_s	G	β_s
α_s	1	-	-
G	0.066	1	-
β_s	-0.138	0.635	1
Panel C	α	G	β
Europe middle income	0.175	0.879	0.743
New EU members	0.143	1.200	0.772
Europe non EU	0.143	1.202	1.142
Europe high-income	0.136	0.894	0.934
Turkey	0.125	0.952	0.752

Notes: We run the shift-share diagnostics suggested by Goldsmith-Pinkham et al. (2020). Panel A shows the share of Rottemberg weights that are positive and negative. Panel B shows the correlation between the Rottemberg weights, the time-shifter shock G, and the just-identified coefficients β. Panel C summarizes the mean of α, G, and β for the top 5 origin regions in terms of weights

95% confidence interval is 0.006) and the point estimate is substantially smaller than the contemporaneous effect (e.g., the contemporaneous effect is approximately 4 times larger).

Table C3: Pre-trend analysis

Lag h	Value	95% Conf. Interval	F stat	N obs
0	0.87	[0.78 , 0.97]	78.74	4382
1	0.22	[0.06 , 0.37]	80.00	3886
2	-0.14	[-0.30 , 0.05]	80.38	3619
3	-0.08	[-0.25 , 0.07]	76.62	3419
4	-0.02	[-0.19 , 0.12]	76.47	3249
5	-0.08	[-0.24 , 0.10]	79.10	3137

Notes: This table reports the 2SLS estimation output for equation 18 for the outcome variable $Ln\left(\frac{\text{Wage Bill Immig}_{j,t-h}}{\text{Wage Bill Natives}_{j,t-h}}\right)$. Confidence intervals are bootstrapped with 5,000 repetitions as described in Section 5.2.

To further assess whether the initial shares used in the instrument correlate with firm covariates that could also affect changes in firm-level outcomes, we perform two additional tests. First, Table C4 shows the importance of key firm characteristics in explaining variation in the initial immigrant share. The firm characteristics that we included are firm size, measured by employment (in logs), the share of exports in sales, the cost-to-sales ratio, the share of college-educated workers, the average immigrant-to-native wage bill ratio, and the average age. This table shows that firm characteristics explain less than 1%

of the variation in the shares, indicating minimal influence from these observables.[55]

Table C4: Correlation between firm initial shares and characteristics

	Initial share 03
Log employment	0.0021
	(0.0019)
Share exports in sales	0.0001
	(0.0001)
Cost-to-sales ratio	-0.0012
	(0.01)
Wage bill-to-cost ratio	-3.07
	(4.96)
Share college Workers	-0.0267
	(0.0218)
Avg immigrant-to-native wage ratio	0.0017
	(0.0128)
Avg age	-0.0001
	(0.0002)
N	6,426
R-sq	**0.001**

Notes: $^{***} = p < 0.01, ^{**} = p < 0.05, ^{*} = p < 0.1$. We regress the firm's initial share on origin region fixed effects and firm characteristics, including total employment (log), export share, cost-to-sales ratio, wage bill share, share of college workers, immigrant-to-native wage ratio, and average workforce age. The key statistic is the R-squared. Results are consistent when not controlling for origin fixed effects.

Second, we test whether our firm-level shift-share instrument is mean independent of the factors that may affect the changes in the outcome variable (Roth et al., 2023). We re-estimate the model adding pre-shock firm characteristics interacted with year dummies and present the estimates in Table C5. All of these regressions include the pre-shock firm characteristics included in the baseline specification. Given the stability of the estimates across specifications, it seems plausible that our estimates are not contaminated by the effects associated with the firm characteristics that affect firm performance.

Table C5: Robustness of ϵ when controlling for covariates

	(1)	(2)	(3)	(4)	(5)
β	0.873***	0.871***	0.867***	0.866***	0.871***
	(0.056)	(0.057)	(0.056)	(0.056)	(0.057)
N	4382	3326	3473	3473	3056
1st stage F stat	78.97	75.16	78.21	78.28	79.91
Control	Baseline	Export share	Employment	College share	Imm-Nat wage

Notes: $^{***} = p < 0.01, ^{**} = p < 0.05, ^{*} = p < 0.1$. We run equation 48 but controlling for specific covariates at their 2003 level interacted with time fixed effects. We include the covariates export share, total firm employment, college share, and the immigrant-to-native wage bill ratio within the firm.

[55] We also find minimal explanatory power of these characteristics if we estimate individual regressions for each of the top 5 countries.

C.4 Estimation of Remaining Structural Parameters

Elasticity of labor supply ν: We estimate ν using information on the variability of wages in our sample. A property of the Fréchet distribution is that it governs both, the dispersion in abilities and the elasticity of labor supply. If abilities across sectors and countries are very dispersed, individuals will not respond as much in their choices when wages change. On the other hand, if ability draws across options are very concentrated, workers' choices will be very sensitive to changes in the wage.

To estimate ν, we follow the literature estimating the labor supply elasticity using properties of extreme value distributions and data of the distribution of wages (e.g., Hsieh et al. (2019), Fan (2019), and Lee (2020)). Specifically, the Fréchet distribution assumption implies that the wage distribution for workers from the same nationality who choose the same sector-country pair has a squared coefficient of variation as in equation 49.

$$\frac{\text{Variance}}{\text{Mean}^2} = \frac{\Gamma\left(1-\frac{2}{\nu}\right)}{\left(\Gamma\left(1-\frac{1}{\nu}\right)\right)^2} - 1 \tag{49}$$

Following this relationship, we can compute the left-hand side from the data to recover the implied value of ν. Note that the elasticity governs both the sector choice for natives and migrants, and the country choice for migrants. Since we only have data on workers living in Germany, our estimation predominantly captures the elasticity of supply across sectors.[56]

To estimate ν with our data, we restrict the sample to German individuals between 25 and 65 years of age who work at the establishments in our sample between 2003 and 2011.[57] We run an individual-level regression of log wages on different sets of fixed effects and take the residual of that regression. We consider three sets of fixed effects: 1) industry-year fixed effects, 2) industry-year fixed effects, district fixed effects, and age and gender controls, and 3) a fully saturated model that includes age-gender-district-industry-year fixed effects. Implicitly, the residual of these regressions is the data counterpart of the ability draws taken by individuals in the model. We then take the exponent of that residual and compute the left-hand side of equation 49. Table C6 shows that our estimates range from 4.90 to 6.14. Our preferred value is $\hat{\nu} = 6.14$, which is from the specification where we add the most stringent fixed effects. Our results welfare effect results are robust to different values of this elasticity, with our baseline estimate yielding the most conservative native workers' welfare gains from immigration. Specifically, the welfare gains for $\nu = 4.90$, $\nu = 5.37$, and $\nu = 6.14$ are 0.10%, 0.14%, and 0.18% respectively.

[56] This is desirable since our main exercise changes the total number of immigrants by a given amount.

[57] We focus on Germans as, in the model, they only choose which sector to work in, which is the main variation of the estimation. Adding immigrants to the calculations gives practically identical results.

Table C6: Estimates for ν

	(1)	(2)	(3)
Coefficient of variation^2	0.104	0.082	0.059
Implied ν	4.90	5.37	6.14
Fixed effects	Industry × year	Industry × year, age, gender, district	Fully saturated

Notes: We run a regression of log wages on a set of fixed effects, and take the residual. We then take the exponent and calculate the squared coefficient of variation. In column 1 we use sector-year fixed effects. In column 2, we use sector-year fixed effects, district fixed effects, and age and gender as controls. In column 3, we use a fully-saturated set of fixed effects at the sector-age-gender-district-year level.

Cobb-Douglas parameter α: We set the $\alpha = 0.68$ to match the share of the tradable and non-tradable sector in the expenditure of German consumers using World Input-Output Tables (WIOT).

Indirect inference approach: As a first step, we proceed to do some normalizations, since not all parameters can be separately identified. The mean fixed costs of hiring immigrants ($\mu_{f,k}$), the mean productivity of immigrants ($A_{o,k}$), and the migration cost ($\phi_{o,\ell,k}$) cannot be separately identified from the immigrant share in the production function (β_k), so we normalize the first one to 0 and the remaining two to 1. We assume the mean productivities in each sector are equal to 1 ($\mu_{\psi,k} = 1$).

As a second step, we proceed to estimate the remaining 14 parameters which we jointly calibrate to minimize the distance between the simulated moments by the model and the observed moments in the data. Table C7 shows the fourteen moments that are targeted in the estimation, their observed values in the data, and the ones generated by the model. The model does a good job in approximating their observed values.

Table C7: Simulated vs data moments

Moment description	Simulated	Data	Moment description	Simulated	Data
Aggregate s_T	0.91	0.91	$\mathbb{E}(1-s_{j,p90}) - \mathbb{E}(1-s_{j,p50}), NT$	0.008	0.008
Aggregate s_{NT}	0.93	0.93	Share of firms hiring immigrants, T	0.62	0.62
$\mathbb{V}ar(log(rev_j)\|s_j, exporter_j)$, T	1.37	1.38	Share of firms hiring immigrants, NT	0.60	0.61
$\mathbb{V}ar(log(rev_j)\|s_j)$, NT	1.29	1.29	GDP per capita RoW to Germany	0.32	0.32
$\mathbb{V}ar((1-s_j)/s_,), T$	1.31	1.39	Share of firms exporting, T	0.37	0.37
$\mathbb{V}ar((1-s_j)/s_j), NT$	1.43	1.58	$\mathbb{E}$(Export to Domestic Rev$_j$), T	0.76	0.79
$\mathbb{E}(1-s_{j,p90}) - \mathbb{E}(1-s_{j,p50}), T$	0.027	0.021	$\mathbb{E}(s_j)$	0.94	0.93

C.5 Model Validation

C.5.1 Estimation details of equation 20:

The parameter θ_1 measures the overall effect of an immigration inflow on firms operating in that labor market and θ_2, our coefficient of interest, captures whether firms of different sizes have differential responses to this inflow. If positive, it implies that a rise in the share of immigrants in a labor market promotes faster growth in y for larger establishments compared to smaller ones. Given that we control for the standalone variables $I_{m,t}$ and $emp_{j,03}$ (absorbed by the firm fixed effect), θ_2 is identified by comparing the differential growth of two firms i and i' in a labor market with a relatively large inflow of immigrants with the differential growth of two firms j and j' with same sizes located in a market with a low inflow of immigrants.[58]

Even though the fixed effects and controls included in the empirical specification aim to capture unobservable shocks and establishment heterogeneity, OLS estimates will be upward-biased if, for example, productivity shocks at the local labor market level improve establishment outcomes and attract migration inflows into the region. To address these endogeneity concerns, we follow an IV approach inspired by Card (2001) and Ottaviano et al. (2018), with the following shift-share instrument:

$$Z_{m,t} = \sum_{o \neq g} \frac{\text{Wage Bill}_{o,m,2003}}{\text{Wage Bill}_{m,2003}} \frac{\Xi_{o,t}}{\Xi_t} \tag{50}$$

where $\text{Wage Bill}_{o,m,2003}$ is the wage bill earned by immigrants from origin country o in labor market m in our initial year 2003. $\text{Wage Bill}_{m,2003}$ is the total wage bill spent in the labor market in 2003 ($\sum_o \text{Wage Bill}_{o,m,2003}$). The initial share is interacted with a time-shifter that captures the national growth rate, from 2003 to year t, of immigrants from origin o relative to the working-age population growth in Germany. The validity of this instrument relies on the assumption that the geographic distribution of immigrants by origin in 2003 is not correlated with local economic conditions in any year t conditional on our controls.

Using revenues and the ratio of immigrant-to-native wage bill as the outcome variable of equation 20, we find that larger firms expand more and become more immigrant-intensive in response to an immigration shock. Table C8 and Table C9 present the OLS and 2SLS estimates, respectively, for revenues (columns 1-3) and the immigrant-to-native wage bill ratio (columns 4-6). We present the results for the full sample as well as for the tradable and non-tradable sectors separately. Table C10 reports the first stage estimation

[58] e.g., $(\Delta y_i - \Delta y_{i'})$ - $(\Delta y_j - \Delta y_{j'})$ where i and i' are in market m, j and j' are in market m', and the difference in size between i and i' is the same than between j and j'.

results, showing that the instruments are strong and that the sign of the estimates is as expected.

Column 1 of Table C9 shows that an increase in the share of immigrants in the labor market increases the revenue of large establishments relative to small establishments.[59] Columns 2 and 3 show that this heterogeneous effect by firm size is driven primarily by establishments in the tradable sector. Establishments in the non-tradable sector do not seem to differentially respond to the immigration shock. Column 4 suggests that immigration into a local labor market has no differential impact on the immigrant intensity of establishments, but once again, this result masks significant heterogeneity across sectors. Column 5 shows that large firms in the tradable sector increase their immigrant intensity relative to small firms, while column 6 shows that this heterogeneous effect across firm size is absent in the non-tradable sector.

Table C8: Effects of immigration for firms of different sizes: OLS estimates

	Log of Revenues			Immigrant-Native Wage Bill		
	(1) All	(2) Tradable	(3) Non-tradable	(4) All	(5) Tradable	(6) Non-tradable
θ_1	-7.838	-11.41*	-2.807	-1.376	-3.771**	0.846
	(5.03)	(6.237)	(5.963)	(0.986)	(1.637)	(1.01)
θ_2	2.844**	3.853**	1.355	0.434**	0.868***	0.0287
	(1.203)	(1.483)	(1.31)	(0.176)	(0.273)	(0.194)
N observations	5,212	2,923	2,289	5,212	2,923	2,289

Notes: $***= p < 0.01, **= p < 0.05, *= p < 0.1$. This table reports the OLS estimation results of equation 20. We restrict the sample to years between 2008 and 2011, and establishments with more than 10 employees. Standard errors are two-way clustered at the industry and labor market level.

[59] Our finding that larger firms benefit more than smaller ones aligns with evidence from other settings where firms actively select immigrants, such as the U.S. firm-sponsoring visa system (Mahajan, 2024). This pattern may not hold in contexts where immigrants were readily available to firms, as in France Mitaritonna et al. (2017) or Israel Arellano-Bover and San (2020).

Table C9: Effects of immigration for firms of different sizes: 2SLS estimates

	Log of Revenues			Immigrant-Native Wage Bill		
	(1) All	(2) Tradable	(3) Non-tradable	(4) All	(5) Tradable	(6) Non-tradable
θ_1	-15.99**	-29.96***	0.425	-2.15	-6.52**	3.65
	(7.79)	(7.98)	(7.30)	(2.04)	(2.75)	(3.58)
θ_2	4.09**	6.93***	0.018	0.54	1.41***	-0.79
	(1.62)	(1.86)	(1.01)	(0.39)	(0.47)	(0.76)
N observations	5,212	2,923	2,289	5,212	2,923	2,289

Notes: $***=p<0.01, **=p<0.05, *=p<0.1$. This table reports the 2SLS estimation results of equation 20. We restrict the sample to years between 2008 and 2011, and establishments with more than 10 employees. Standard errors are two-way clustered at the industry and labor market level.

Table C10: Effects of immigration for firms of different sizes: First stage regressions

	Full sample		Tradable sector		Non-Tradable sector	
	$I_{m,t}$	$I_{m,t} \times log(\text{size}_{03})$	$I_{m,t}$	$I_{m,t} \times log(\text{size}_{03})$	$I_{m,t}$	$I_{m,t} \times log(\text{size}_{03})$
$Z_{m,t}$	2.80***	2.16	2.87***	2.31	2.95***	3.01
	(0.54)	(1.71)	(0.61)	(2.39)	(0.61)	(1.58)
$Z_{m,t} \times log(\text{size}_{03})$	0.03	2.43***	0.05	2.63***	-0.06	1.84***
	(0.04)	(0.37)	(0.05)	(0.42)	(0.06)	(0.44)
N	5,212		2,923		2,289	
Kleinberg-Paap F-stat	20.45		22.49		10.50	

Notes: $***=p<0.01, **=p<0.05, *=p<0.1$. This table reports the first-stage estimation results corresponding to equation 20. The Kleibergen-Paap F-stat tests for the joint significance of both instruments. The first two columns are the first stages for the full sample, columns 3 and 4 restrict the sample to establishments in the tradable sector, and columns 5 and 6 to the non-tradable sector.

Test of the identification assumptions: We follow the suggestions in Goldsmith-Pinkham et al. (2020) and Borusyak et al. (2021) and test for pre-trends in Table C11. These estimates suggest that we can not reject the hypothesis that all coefficients are statistically different from zero at a 5% confidence level.

Robustness of our estimates to alternative sets of controls: Table C12 evaluates the sensitivity of our estimates to different controls. Column 2 removes the firm size-specific trend, column 3 removes the cost-to-revenue control, column 4 removes the industry-time fixed effects, column 5 removes the local labor market trends, and column 6 includes labor market-year fixed effects.

Table C11: Pre-trends tests

	$t-1$	$t-2$	$t-3$	$t-4$	$t-5$
Log Revenues					
θ_1	-14.41*	1.558	-8.857	-0.667	3.994
	(7.601)	(6.313)	(7.814)	(6.218)	(5.022)
θ_2	3.637*	0.649	1.657	-1.144	-0.227
	(1.865)	(1.138)	(2.073)	(1.116)	(0.961)
Immigrant-to-native wage bill					
θ_1	0.265	-0.504	-0.411	-2.254	-1.414
	(1.978)	(1.856)	(1.577)	(2.891)	(2.364)
θ_2	0.136	0.21	0.228	0.447	0.332
	(0.374)	(0.303)	(0.294)	(0.503)	(0.415)
N	5208	5203	5203	5205	5209

Notes: $^{***} = p < 0.01, ** = p < 0.05, * = p < 0.1$. The dependent variable is log revenues in the top panel and the ratio of immigrant to native wage bill in the bottom panel. In each column, we take lags of the dependent variable, going from one-year lag to five-year lag. We control for establishment fixed effects, 2-digit industry-time fixed effects, local labor market time trends, log employment of the firm in 2003 interacted with year fixed effects, and the wage bill of the firm in 2003 relative to revenues interacted with year fixed effects. Standard errors are clustered at the industry-labor market level. The sample is restricted to establishments with more than 10 employees.

Table C12: Robustness exercises for main specification

	Baseline	No employment control	No cost ratio control	No industry-year fixed effects	No local labor market trend	labor market - year fixed effects
Log Revenues						
θ_1	-15.99**	-17.16**	-17.42**	-28.31***	-8.166	-
	(7.794)	(7.745)	(8.254)	(8.739)	(5.580)	-
θ_2	4.095**	4.371***	4.428**	7.183***	2.349*	4.265**
	(1.616)	(1.587)	(1.695)	(1.886)	(1.326)	(1.657)
Ratio of Immigrant to Native Wage Bill						
θ_1	-2.154	-2.517	-2.584	-2.014	-3.826*	-
	(2.045)	(2.118)	(2.182)	(2.118)	(1.937)	-
θ_2	0.54	0.626	0.636	0.533	0.810*	0.23
	(0.388)	(0.410)	(0.423)	(0.414)	(0.404)	(0.442)
N	5212	5212	5212	5212	5212	5177
1st stage F-stat	20.48	22.68	22.38	22.57	57.76	52.87

Notes: $^{***} = p < 0.01, ** = p < 0.05, * = p < 0.1$. The dependent variable is log revenues in the top panel and the ratio of immigrant to native wage bill in the bottom panel. We control for establishment fixed effects, 2-digit industry-time fixed effects, local labor market time trends, log employment of the firm in 2003 interacted with year fixed effects and the wage bill of the firm in 2003 relative to revenues interacted with year fixed effects. Standard errors are clustered at the industry-labor market level. The sample is restricted to establishments with more than 10 employees. Column 1 shows the baseline results. Column 2 removes the firm employment control, column 3 removes the cost-to-revenue control, column 4 removes the industry-time fixed effects, column 5 removes the local labor market trends, and column 6 includes labor market-year fixed effects.

C.5.2 Empirical estimates of $\epsilon^y_{j,t}$:

Table C13 presents the average value of $\epsilon^y_{j,t}$ from equation 21 by firm size and sector, which is used to compare the responses implied by our quantitative model. In the tradable sector, a 1% increase in the labor market immigrant share decreases establishments' revenues in the lowest size decile by 0.71 log-points on average, while increasing estab-

lishments' revenues in the highest decile by 1.12 log-points on average. The elasticity of revenues in the non-tradable sector, on the other hand, is similar across establishments of different sizes.

We find a similar pattern in each sector when looking at the response of the relative wage bill between immigrants and natives across size deciles. In the tradable sector, a 1% increase in the share of immigrants in the labor market would decrease the ratio of an establishment in the lowest decile by 0.17 while increasing the ratio for an establishment in the highest decile by 0.19. The elasticities across deciles in the non-tradable sector seem to be decreasing with size, but are not statistically significant.

Table C13: Estimated response to immigration by firm size decile

	Size deciles									
Tradable	1	2	3	4	5	6	7	8	9	10
Revenues	-0.71	-0.47	-0.36	-0.28	-0.21	-0.06	0.11	0.34	0.52	1.12
Relative Immigrant WB	-0.17	-0.12	-0.09	-0.08	-0.07	-0.03	0.00	0.04	0.08	0.19
Non-tradable	1	2	3	4	5	6	7	8	9	10
Revenues	0.03	0.03	0.03	0.02	0.02	0.03	0.03	0.02	0.03	0.03
Relative Immigrant WB	0.12	0.08	0.08	0.05	0.03	0.03	0.01	-0.01	-0.03	-0.07

Notes: We use estimates from Table C9 to compute the $\epsilon^{y}_{j,t}$ for each observation using equation 21. We divide establishments into deciles based on employment and, for each decile and sector, compute the mean value of $\epsilon^{y}_{j,t}$.

D Quantitative Appendix

D.1 Sector-level Adjustment to the Inflow of Immigrants

Appendix Table D1 shows the sectoral effects on employment and wages in terms of labor units (i.e., number of workers) and effective units. The influx of immigrants decreases the relative wage between immigrants and natives, and both sectors become more immigrant-intensive. As they become more competitive, both sectors expand their production and total employment in terms of effective units. The rise in immigrant intensity and production is associated with increased demand for natives in the non-tradable sector *relative* to the tradable sector, leading to a reallocation of native-born workers from the tradable to the non-tradable sector.

Table D1: Effect of immigration on employment and wages

	Labor units		Effective units	
Employment	Tradable	Non-Tradable	Tradable	Non-Tradable
Total	3.94%	3.29%	6.10%	5.31%
Native	-0.09%	0.20%	-0.08%	0.17%
Immigrant	20.00%	20.00%	16.49%	16.49%
Wages				
Natives	-0.06%	-0.06%	-0.07%	-0.03%
Immigrants	-3.00%	-2.95%	-2.63%	-2.10%

Notes: We compute the changes on endogenous variables of going from the observed equilibrium to an equilibrium where the number of immigrants is 20% higher. The wage in the RoW, w_r, is the numeraire.

D.2 Comparing Policies that Affect Firm Heterogeneity

This section studies policies that alter the hiring costs of immigrants for specific firms. In Section 3.3, we mentioned that the German government implemented policies to reduce the frictions faced by small and medium enterprises (SMEs) for recruiting immigrants. These policies motivate our first counterfactual analysis, referred to as "Policy 1". We divide firms into two groups based on whether their productivity ψ_j is above or below the median. The policy lowers the immigrant recruitment cost f_j for below-median productivity firms to the 10th percentile of the f_j distribution. We compare it with three alternative policies to shed light on how policies that alter the distribution of firms' immigrant intensities can affect aggregate outcomes. An alternative policy reduces frictions by the same total amount but for firms in the top half of the productivity distribution ("Policy 2"). This policy aligns with efforts to facilitate hiring by large firms, such as the U.S. L-1 visa program, which allows multinational companies to transfer employees across subsidiaries and streamline applications for multiple workers in the same role. While Policies 1 and 2 are not sector-specific, they primarily affect non-tradable firms because they face higher frictions for hiring immigrants. This motivates two additional policies: reducing frictions in the tradable sector by the same total amount, either for firms below the sector's median productivity ("Policy 3") or for those above it ("Policy 4").

We begin by studying Policy 1's effect on aggregate outcomes (Column 1 of Table D2). Appendix Figure D1a plots the reallocation of immigrants and native workers across sectors and productivity deciles. The policy substantially raises immigrant employment in below-median productivity firms within the non-tradable sector, while the increase in

the tradable sector is modest. This difference across sectors arises for two main reasons. First, non-tradable firms face higher initial costs f_j than firms in the tradable sector, so reducing these costs to the 10th percentile represents a larger relative shock. Second, low-productivity firms in the tradable sector hold a much smaller market share within their industry.[60] As a result, even with lower fixed costs, unproductive firms in the tradable sector remain at a steep disadvantage relative to their domestic competitors and see modest expansion in hiring or production. The increase in immigrant hiring by small firms in the non-tradable sector crowds out native coworkers, who relocate to higher productivity firms in both sectors. Regarding the aggregate effects of the policy, real earnings for native workers increase but only slightly due to countervailing effects. On one hand, immigrant employment increases, which pushes prices down for all consumers by 0.15%. On the other hand, the influx of immigrants reduces native-born wages due to higher labor market competition in the non-tradable sector. Natives mitigate such losses by moving to the tradable sector, but overall earnings decrease by 0.11%. While native-born workers' earnings decline, firm owners' income rises due to higher profits, leading to a net total income increase of 0.32% for Germans.

We now turn to the analysis of alternative policies, highlighting how aggregate outcomes differ relative to Policy 1. Lowering f_j for firms at the top of the productivity distribution (Policy 2) primarily increases immigrant employment by top-productivity firms in the non-tradable (see Appendix Figure D1b). The aggregate effects more than double those of Policy 1 for two reasons. First, the total inflow of immigrants is more than double that under Policy 1 because the affected firms are larger and their production and immigrant labor demand is more elastic to drops in immigrant hiring costs. Second, larger firms account for a larger share of the consumption basket of consumers, so a drop in the price of their goods induces a stronger drop in the price index of consumers. Lowering fixed costs for firms in the tradable sector below the median productivity (Policy 3) has no better effects on the aggregate economy than Policy 1. This policy is not effective because, as mentioned before, a drop in hiring costs is not enough for the below-median productivity firms to overcome productivity differences within the sector and expand (see Appendix Figure D1c for details of labor adjustment). Finally, lowering fixed costs for high-productivity firms in the tradable sector (Policy 4) yields better outcomes for native workers and overall welfare. Large tradable firms face more elastic demand, and can expand production without significantly lowering prices. This implies that when high-productivity firms in the tradable sector increase immigrant employment (see Appendix Figure D1d), they do not reduce the prices of other factors as much as firms in the non-tradable sector would.

[60] For reference, while the top decile of tradable firms accounts for 90% of production in that sector, the top decile in the non-tradable sector produces only 30%.

Table D2: Aggregate effects of alternative immigration policies

	Policy 1	Policy 2	Policy 3	Policy 4
Total income	0.32%	0.83%	0.02%	1.95%
Earnings Natives	-0.11%	-0.24%	0.00%	0.03%
Price Index	-0.15%	-0.31%	-0.01%	-0.08%
Real Earnings Natives	0.04%	0.07%	0.00%	0.11%
Real Wages per efficiency units				
Natives - T	0.09%	0.18%	0.00%	-0.02%
Natives - NT	-0.08%	-0.17%	0.01%	0.40%
Immigrants - T	0.14%	0.62%	0.07%	4.26%
Immigrants - NT	2.52%	5.06%	0.01%	0.38%
Employment				
Natives - T	0.32%	0.65%	-0.01%	-0.80%
Natives - NT	-0.72%	-1.46%	0.03%	1.80%
Immigrants - T	0.84%	3.81%	0.43%	29.02%
Immigrants - NT	16.50%	35.28%	0.04%	2.18%
Immigrants	4.36%	10.42%	0.26%	17.56%

Notes: The table shows the change in endogenous variables of four counterfactual reductions in f_j. Policy 1 reduces f_j to the firms with productivity ψ below the median of the economy. This fixed cost is lowered to the 10th percentile of the distribution. The three alternative policies reduce fixed costs by the same total amount but for firms in the top half of the productivity distribution (Policy 2), for the bottom half of firms in the tradable sector (Policy 3), and for the top half of firms in the tradable sector (Policy 4).

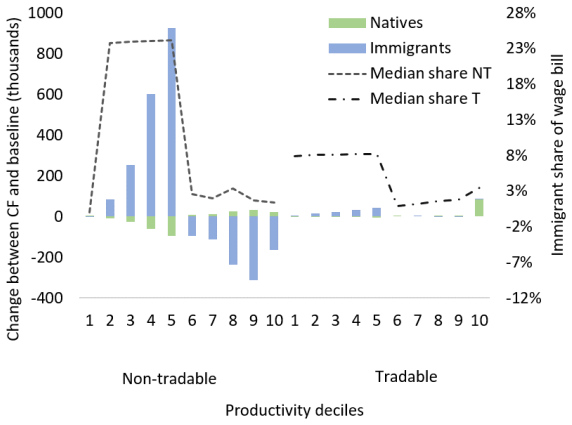
Overall, our findings suggest that policies that target highly productive firms in tradable sectors might have the best outcomes for both labor and total income. Also note that this policy increases the number of immigrants by 17.56%, a smaller inflow than in our baseline counterfactual, yet it leads to a larger rise in welfare. Thus, firm-targeted immigration policies that reduce fixed costs to hire immigrants are a rich policy tool capable of achieving better aggregate outcomes than policies focused solely on inflow size.

D.3 Eliminating Fixed Cost f_k^{imm}

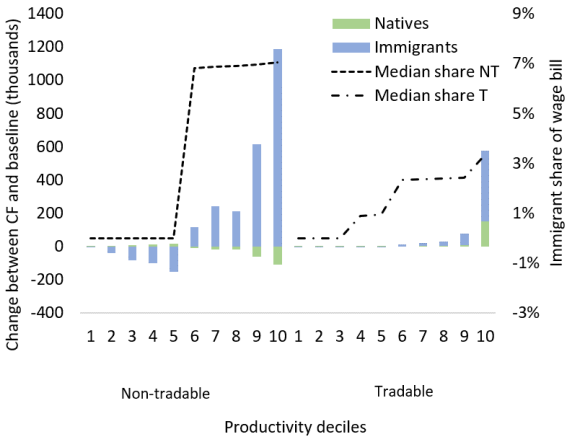
This section studies how the aggregate outcomes change if we remove the initial cost of hiring immigrants, $f_T^{imm} = f_{NT}^{imm} = 0$. This counterfactual is in line with a policy like the restrictions lifted by the EU enlargement for immigrants from all countries. As shown in Figure D2, there is a large increase in the number of immigrants hired by small firms in the non-tradable sector. Natives in these firms experience higher competition and lower wages as shown in Table D3, pushing them to move to large firms in the tradable sector. Smaller firms in the tradable sector do not expand as much, since such firms have a very small market share to begin with, so the lower frictions are not enough to help them take market share from larger productivity firms in the sector. The aggregate real-wage effects of this policy are close to zero, as the reduction in prices is almost identical to the

Figure D1: Immigrant and Native employment - alternative policies.

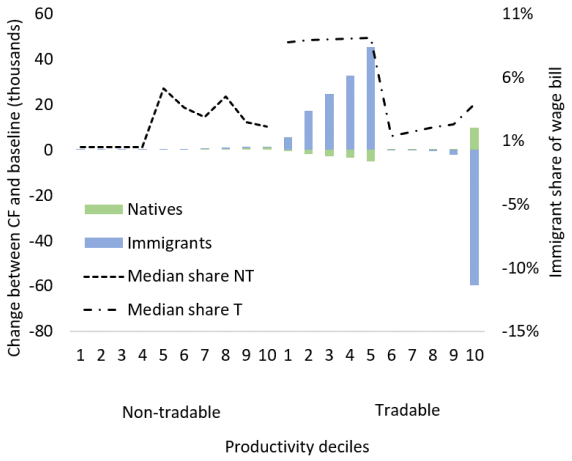
(a) Policy 1



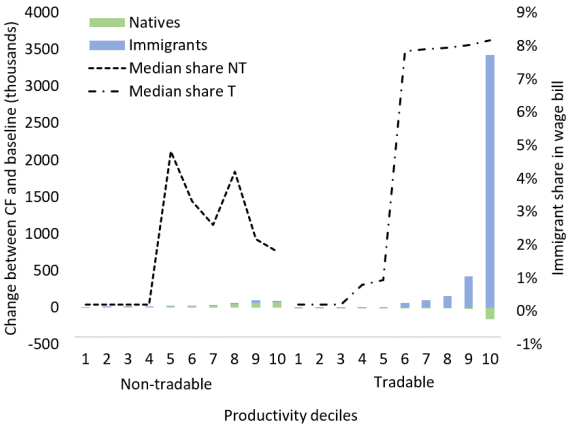
(b) Policy 2



(c) Policy 3



(d) Policy 4



Notes: The x-axis shows the distribution of firms in the non-tradable (left) and tradable (right) sectors in terms of deciles of productivity. The bars plot the absolute change in thousands of workers employed in each decile between the counterfactual scenario and the baseline. The dotted lines plot the median immigrant share under the counterfactual scenario. Each figure four counterfactual reductions in f_j. Policy 1 reduces f_j to the firms with productivity ψ below the median country of the country. This fixed cost is lowered to the 10th percentile of the distribution. The three alternative policies reduce fixed costs by the same total amount, but for firms in the top half of the productivity distribution (Policy 2), for the bottom half of firms in the tradable sector (Policy 3), and for the top half of firms in the tradable sector (Policy 4).

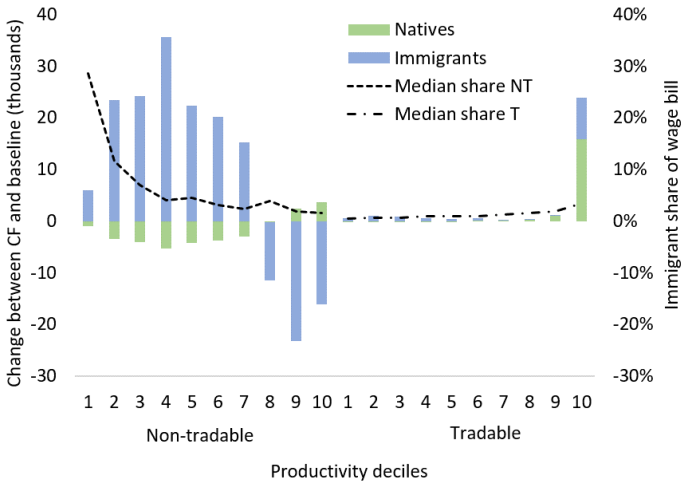
reduction in wages.

Table D3: Aggregate effects of policy eliminating the initial fixed cost of hiring

	Removing cost to begin hiring immigrants
Earnings Natives	-0.02%
Price Index	-0.02%
Real Earnings Natives	0.00%
Real Wages	
Natives - T	0.01%
Natives - NT	-0.02%
Immigrants - T	0.01%
Immigrants - NT	0.27%
Employment	
Natives - T	0.07%
Natives - NT	-0.15%
Immigrants - T	0.08%
Immigrants - NT	1.67%

Notes: We compute the changes on the key endogenous variables of going from the observed equilibrium to an equilibrium where we implement the policy of setting the initial fixed cost of hiring immigrants to zero for all firms.

Figure D2: Immigrant and Native employment - alternative policies.



Notes: The x-axis shows the distribution of firms in the non-tradable (left) and tradable (right) sectors in terms of deciles of productivity. The bars plot the absolute change in thousands of workers employed in each decile between the counterfactual scenario and the baseline. The dotted lines plot the median immigrant share under the counterfactual scenario. The counterfactual policy is setting the initial cost of hiring immigrants as zero to all firms ($f_T^{imm} = f_{NT}^{imm} = 0$).